Timing the Factor Zoo *

Andreas Neuhierl† Otto Randl‡
Christoph Reschenhofer§ Josef Zechner¶

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Abstract

We provide a comprehensive analysis of the timing success for equity risk factors. Our analysis covers over 300 risk factors (factor zoo) and a high-dimensional set of predictors. The performance of almost all groups of factors can be improved through timing, with improvements being highest for profitability and value factors. Past factor returns and volatility stand out as the most successful individual predictors of factor returns. However, both are dominated by aggregating many predictors using partial least squares. The median improvement of a timed vs. untimed factor is about 2% p.a. A timed multifactor portfolio leads to an 8.6% increase in annualized return relative to its naively timed counterpart.

Keywords: time-varying risk premia, factor investing, partial least squares
JEL codes: G10, G12, G14

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†Olin Business School, Washington University in St. Louis, andreas.neuhierl@wustl.edu
‡WU Vienna University of Economics and Business, otto.randl@wu.ac.at
§WU Vienna University of Economics and Business, christoph.reschenhofer@wu.ac.at
¶WU Vienna University of Economics and Business, josef.zechner@wu.ac.at

1 Introduction

Empirical asset pricing research has identified a staggering quantity of priced risk factors. While it may be challenging to rationalize all these factors as independent sources of systematic risk, it is clear that one needs a multifactor model to explain the cross-section of asset returns. In light of the empirical asset pricing literature, it is also uncontroversial that risk premia vary conditionally over time. At the market level, for example, Fama and French (1988) find that returns are predictable by the dividend-price ratio. This opens the arena for market timing, but, in a multifactor world, the more general question concerns the timing of all sources of systematic risk – factor timing. Given the plethora of factors, it is no surprise that a large number of time series predictors for their returns has also been suggested in the literature. The combination of the large numbers of factors and predictors amplifies the empirical challenge in answering the question: *Are factor premia predictable?* We carry out a comprehensive analysis using over 300 factors and 39 signals and find that factor timing is indeed possible and profitable. We thereby resolve conflicting findings in the academic literature that result from choosing a smaller subset of factors and/or predictors.

We first establish a benchmark and study the benefits from factor timing in a univariate fashion, i.e. we forecast each factor using each of the 39 signals and then aggregate over the signal class. The analysis reveals that versions of momentum and volatility signals are able to provide improvements on a broad basis. Other signal classes (valuation spreads, characteristic spreads, reversal and issuer-purchaser spread) provide improvements, but the results vary more strongly depending on whether we study improvements in raw returns, alphas or Sharpe ratios. Next, we aim to improve the univariate analysis by aggregating the signals. Many of the predictive signals are highly correlated as they aim to capture the same phenomenon, such as versions of momentum. Since conventional ordinary least squares regression is known to perform rather poorly in such settings, we resort to dimension-reduction techniques to obtain a low dimensional representation of the predictive information. We use partial least squares regression, which provides a data-driven method to aggregate the signals for each factor. However, our setup allows for heterogeneous dynamics across factors. Partial least squares leads to improvements in statistical and economic terms. For the median factor, we achieve an out-of-sample R^2 of 0.76% and an improvement of annual returns of approximately 2 percentage points. We correctly forecast the sign of a factor return 57% of the time and most notably the improvements relative to passive buy-and-hold are not confined to a small part of the sample, but accrue almost equally over the full sample.

We also study the benefits of factor timing for multifactor portfolios. We build quintile portfolios of factors, i.e. we go long the factors for which we forecast the highest returns and short the factors for which we forecast the lowest returns. The resulting "high-low" portfolio achieves an annualized Sharpe ratio of 1.23. This is a significant improvement over merely sorting factors on their historical mean returns, which leads to an annual Sharpe ratio of 0.79.

It is well known that factor premia decline in their post-publication period (McLean and Pontiff, 2016). We also find a pronounced decline for the average factor post publication of about 3% in average annual returns. However, we find that the return difference between timed and untimed factors remains nearly constant in the pre- vs. post publication periods. Put differently, the returns to factor investing decay, but factor timing remains successful.

While previous research on factor timing has taken the factors as primitives, we also look under the hood and study the portfolio composition of optimal factor timing portfolios. This bottom-up approach allows us to answer important questions about the properties of timing portfolios such as turnover as well as their style tilts. This approach also allows us to focus on large stocks and long-only portfolios. We find that long-only timing portfolios that focus on large stocks exhibit moderate levels of turnover and could likely be implemented in practice. The large-cap timing portfolios achieve an annual average return of approximately 13.5%, whereas a value weighted portfolio of these stocks only averages 9.3% p.a. over the same period. Nonetheless, the optimal large-cap timing portfolio still contains almost 400 stocks on average, thereby providing sufficient diversification of idiosyncratic risk. The magnitude of the outperformance is persistent across periods of high and low market risk (measured using the VIX) and during NBER recessions and expansions.

The early literature on factor timing is largely concerned with the market index. While the overall literature on market timing is too large to be summarized here, we refer to the important early contributions of Shiller (1981) and Fama and French (1988). Their early work has been extended to other style factors, such as value by Asness, Friedman, Krail, and Liew (2000) and Cohen, Polk, and Vuolteenaho (2003), who show that the expected return on a value-minus-growth strategy is atypically high at times when its spread in the book-to-market ratio is wide. More recently, Baba Yara, Boons, and Tamoni (2021), show returns for value strategies in individual equities, commodities, currencies, global bonds and stock indexes are predictable by the value spread between stocks ranked in the top percentiles versus those in the bottom.

An important methodological innovation is due to Kelly and Pruitt (2013), who link disaggregated valuation ratios and aggregate market expectations to document high out-of-sample return predictability for value, size, momentum and industry portfolios. Their finding is particularly useful for our setting as we also need to aggregate many predictors to forecast individual time series. Other approaches to aggregate signals are proposed in Leippold and Rueegg (2019), who use momentum in the weights of an integrated scoring approach to form long-only portfolios that outperform. Dichtl, Drobetz, Lohre, Rother, and Vosskamp (2019) use cross-sectional information about factor characteristics to tilt factors and show that the model loads positively on factors with short-term momentum, but avoids factors that exhibit crowding.

Factor volatility as a potential timing signal deserves special mention as it is subject to controversy. In Brandt, Santa-Clara, and Valkanov (2009)'s parametric portfolio framework, DeMiguel, Martın-Utrera, and Uppal (2024) show that a conditional mean-variance optimal multifactor portfolio whose weights on each factor vary with market volatility outperforms out-of-sample. Moreira and Muir (2017) show that past factor volatility, estimated from past daily returns, is a useful conditioning variable to choose time-varying exposure to individual factors, in particular the market factor. Cederburg, O'Doherty, Wang, and Yan (2020) find that the performance benefits of volatility management disappear once more realistic assumptions are made regarding portfolio implementation, such as trading costs. They conclude that, once such frictions are considered, volatility-managed portfolios exhibit lower certainty equivalent returns and Sharpe ratios than do simple investments in the original, unmanaged portfolios. Barroso and Detzel (2021) consider volatility-managed factor portfolios, applying various cost-mitigation strategies. They find that even in this case, realistic estimates of transactions costs render volatility management unprofitable for all factors, except for the market. Reschenhofer and Zechner (2024) show that portfolio performance can be improved significantly when jointly using volatilities of past factor returns and option-implied market volatilities to determine factor exposures. This multivariate volatility-based factor timing leads to larger improvements when option-implied market returns are right-skewed and exhibit high volatility.

Various implementations of factor momentum have also received considerable attention in the literature. Ehsani and Linnainmaa (2022) show that factor momentum is a likely underlying driver of different forms of classic cross-sectional momentum. Arnott, Kalesnik, and Linnainmaa (2023) show that factor momentum is also the source of industry momentum. Gupta and Kelly (2019) also provide evidence of factor momentum in many popular asset pricing factors. In contrast, Leippold and Yang (2021) argue that factor momentum can largely be attributed to high unconditional rather than conditional returns. Bessembinder, Burt, and Hrdlicka (2024) find significant time-variation in factor premia and argue that many more factors should be considered for pricing.

Haddad, Kozak, and Santosh (2020) extract principal components from 50 popular anomaly portfolios and find significant benefits from factor timing based on the book-to-market ratio. They also discuss broader asset pricing implications of their findings. In particular, they document that a stochastic discount factor (SDF) that takes into account timing information is more volatile and has different time series behavior compared to static alternatives, thereby posing new challenges for theories that aim to explain the cross-section of expected returns. Kelly, Malamud, and Pedersen (2023) allow for cross-predictability; they use signals of all securities to predict each security's individual return. They apply a singular value decomposition to summarize the joint dynamics of signals and returns into "principal portfolios". Using a large sample of equity factors and trading signals, they find factor timing strategies based on principal portfolios to perform well overall and across the majority of signals, outperforming the approach of Haddad et al. (2020). Kagkadis, Nolte, Nolte, and Vasilas (2024) extend Haddad et al. (2020) and incorporate a larger set of observable characteristics, which are independently identified for each principal component portfolio.

Asness (2016) finds timing strategies that are simply based on the "value" of factors to be very weak historically. Asness, Chandra, Ilmanen, and Israel (2017) look at the general efficacy of value spreads in predicting future factor returns. At first, timing based on valuation ratios seems promising, yet when the authors implement value timing in a multi-style framework that already includes value, they find somewhat disappointing results. They conclude that value timing of factors is too correlated with the value factor itself. Adding further value exposure this way is dominated by an explicit risk-targeted allocation to the value factor. Lee (2017) suggests investors are better off focusing on the underlying rationale of risk premia rather than attempting to time factors. Ilmanen, Israel, Moskowitz, Thapar, and Lee (2021) examine four prominent factors across six asset classes over a century. They find only modest predictability, which could only be exploited in a profitable way for factor timing strategies if trading costs are minimal.

This paper differs from the studies discussed above, as we consider a comprehensive set of both factors and timing signals, show how the timing signals can be successfully aggregated, and characterize the resulting long-only portfolios of individual stocks.

2 Data

2.1 Factors

Cross-sectional asset pricing has taken a long journey from single-factor models (e.g., Sharpe, 1964) via parsimonious multi-factor models (e.g., Fama and French, 1992) towards a heavily criticized factor zoo (e.g., Cochrane, 2011; Harvey, Liu, and Zhu, 2016). For many factors, their validity in light of out-of-sample evidence on the one hand and in light of mere replicability on the other hand has come under scrutiny. Chen and Zimmermann (2022) give a positive assessment of preceding academic work. In a massive and open source code replication effort, they reproduce 318 firm-level characteristics. They confirm the original papers' evidence for all but three characteristics and confirm previous findings of performance decaying, but often staying positive out-of-sample.[1] To analyze factor timing, we need a clean data set of factor portfolios that ideally are associated with positive unconditional risk premia, but time variation in returns. Thus, we start from a set of factor portfolios obtained by applying the methodology of Chen and Zimmermann (2022), i.e. rather than imposing our own biases on factor construction and selection, we regard the factor universe as exogenous.

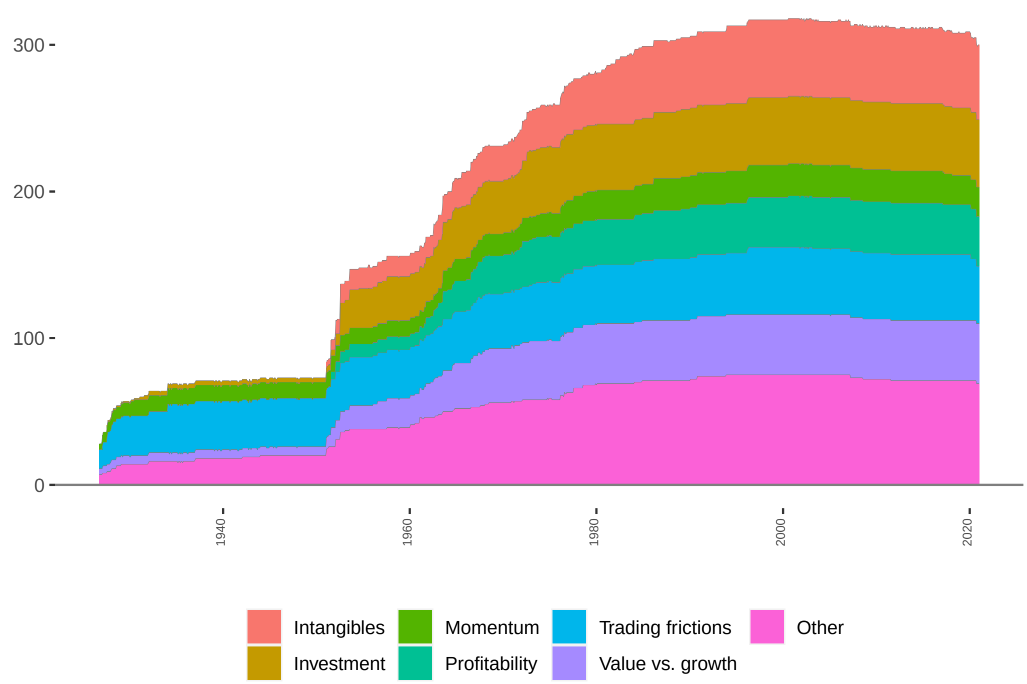
To sort stocks into portfolios, we construct firm characteristics based on data obtained from CRSP, Compustat, IBES, and FRED. Several characteristics require specific data to reconstruct the results of the original studies, and are readily available on the authors' websites. For each characteristic, we follow Chen and Zimmermann (2022) and replicate portfolios defined in the original paper that introduced the anomaly in the literature. We group similar factors based on their economic interpretation. For factors included in Hou, Xue, and Zhang (2020), we follow their classification. For the remaining factors, we group them into the categories intangibles, investment, momentum, profitability, trading frictions, value vs. growth, and other. Our sample covers the time period from 1926 to 2020. Data availability translates into different starting points for the various characteristics and consequently into

[1] Their positive assessment is reinforced by the findings of another open-source project, Jensen, Kelly, and Pedersen (2023).

different starting points for the factor portfolios. In general, price-based characteristics have the longest history, with accounting data and analyst forecasts becoming available later in time. Figure 1 plots the number of factors per category over time. Table A.1 provides detailed information on the characteristics, the original studies, and classification into economic categories. Table A.2 provides descriptive statistics of factor category and individual factor returns.

Figure 1: Number of Factors per Category

This figure shows the number of factors over time. We follow Chen and Zimmermann (2022) and replicate portfolios defined in the original paper that introduced the anomaly in the literature. We group factor portfolios into six economic categories based on the firm characteristics used to construct them: Intangibles, Investment, Momentum, Profitability, Trading frictions, Value vs. growth, and Other. For factors included in Hou et al. (2020), we follow their classification. Table A.1 provides a description of each individual factor and the assigned factor category.



2.2 Timing Signals

We use a broad set of timing signals that have been proposed in the literature and group them into six classes: momentum, volatility, valuation spread, characteristics spread, issuer-purchaser spread, and reversal. We first provide a short summary of the different signals (full details are given in Appendix B).

Momentum: Momentum signals are based on the observation that past factor returns over fairly recent periods positively predict future returns. While the classic definition for momentum is cross-sectional and thus less suited for factor timing, we use variations of time series momentum to construct signals. The simplest variants of momentum-based timing signals rely on the sign of prior returns. Thus, we derive momentum signals that assign a weight of $w_{i,t} = \pm 1$, conditional on the sign of the past factor return over an n-months horizon. We use look-back periods n equal to 1, 3, 6, and 12 months. Ehsani and Linnainmaa (2022) measure the profitability of factor momentum by taking long and short positions in factors based on prior returns. In further variants of timing signals, we follow Gupta and Kelly (2019), and obtain the weights $w_{i,t}$ of the timed factor portfolios as factor i's n-months past return, scaled by m-months past return volatility. Different values for n and m result in different timing signals. Ehsani and Linnainmaa (2022) measure the profitability of factor momentum by taking long or short positions in factors based on prior returns. Thus, we derive momentum signals that assign a weight of $w_{i,t} = \pm 1$, conditional on the sign of the past factor return over an n-months horizon. Finally, we follow Moskowitz, Ooi, and Pedersen (2012) and scale positions such that the timed factor has an ex ante volatility of 40%. In total, we use 16 momentum signals.

Volatility: Moreira and Muir (2017) show that realized volatility predicts future volatility but not returns. Investment strategies that condition factor exposure on recent realized volatility tend to outperform in a risk-adjusted metric. Mirroring the measures analyzed in their paper, we use the realized standard deviation and the variance of daily factor returns over the preceding month to construct timing signals. In a variant, we obtain the variance predictor from an AR(1) process fitted to log variance. Following Cederburg et al. (2020), we estimate another variant where realized variance is calculated as the sum of squared daily factor returns, scaled by the ratio of 22 to the number of trading days in a given month.An additional volatility signal is obtained from volatility of market returns instead of factor returns (DeMiguel et al., 2024). Finally, we follow Reschenhofer and Zechner (2024), who find improved predictability when complementing moments estimated from historical data with option-implied information. We thus use the CBOE VIX index and the CBOE SKEW index for signal construction. The different methods result in a total of seven volatility signals.

Valuation spread: Stock market valuation ratios are a traditional predictor of aggregate returns, (see, e.g., Campbell and Shiller, 1988). Prices scaled by fundamental variables such as dividends, earnings, or book values contain information about expected returns of the market. If the aggregate valuation level predicts aggregate returns, it seems plausible that the relative valuation of value versus growth stocks should predict their relative returns. The value spread – the book-to-market ratio of value stocks minus that of growth stocks – predicts the HML factor return. Similarly, Haddad et al. (2020) use a portfolio's net book-to-market ratio (defined as the difference between the log book-to-market ratio of the long and the short legs) to predict its return. We define value signals similarly, standardizing a factor portfolio's value spread using the rolling and expanding means, respectively. Variants for the value spread differ with respect to the timing of the signals, with variants (i) end of year book and market values, (ii) end of year book value and most recent market value, and (iii) quarterly book and market values. In total, we derive six versions of valuation signals.

Characteristics spread: The unconditional factor portfolios result from sorting individual stocks on a specific characteristic. As noted by Huang, Liu, Ma, and Osiol (2010), it is thus intuitive that the spread in the (sorting) characteristic between the top and the bottom deciles proxies for future return dispersion. To construct the factor-specific characteristic spread, we calculate the difference in the characteristic of the long minus the short leg, and scale the demeaned spread by its standard deviation. We obtain two signal variants, from using a rolling or an expanding mean.

Reversal: Moskowitz et al. (2012) document time series momentum at horizons up to 12 months and reversal for longer horizons. We first compute 60 (120) months past returns and obtain two version of reversal signals: The 60 (120) month reversal signal translates into a weight $w = 1 -$ annualized factor return over the past 60 (120) months.

Issuer-purchaser spread: External financing activities such as equity issuance net of repurchases and debt issuance are negatively related to future stock returns (Bradshaw, Richardson, and Sloan, 2006; Pontiff and Woodgate, 2008). Greenwood and Hanson (2012) find that determining which types of firms issue stocks in a given year helps forecasting returns of factor portfolios. In particular, the differences between firms who recently issued vs. repurchased shares predict returns to long–short factor portfolios associated with those characteristics. We construct issuer-purchaser spreads based on three variants for the determination of net issuance: the difference between sales and repurchase of common stock, the change in split-adjusted shares outstanding, and the change in split-adjusted common shares outstanding. The time series are demeaned using rolling or expanding means, and scaled by standard deviation, resulting in 6 signals.

3 Empirical Analysis

3.1 Univariate Factor Timing

For univariate factor timing, we construct timed factors as versions of the original factor portfolios, using one specific timing signal to scale the returns. More precisely, we obtain

$$f_{i,t+1}^{j} = w_{i,t}^{j} f_{i,t+1} \, , \tag{1}$$

where $f_{i,t+1}^{j}$ is the excess return of the signal-j-timed factor i from time t to $t+1$, $f_{i,t+1}$ is the excess return of the original factor portfolio, and $w^j_{i,t}$ is the timing weight constructed from signal j. We time each one of the $i \in \{1, \dots, 318\}$ factors at monthly frequency, using $j \in \{1, \dots, 39\}$ signals, resulting in 12,402 timed factor portfolios.

3.1.1 Timing Performance for Different Types of Signals

To evaluate the success of factor timing, we first calculate the difference in returns:

$$\Delta \bar{R}_{i,j} = \frac{1}{T} \sum_{t=1}^{T} \left(f^j_{i,t+1} - f_{i,t+1} \right). \tag{2}$$

Second, to incorporate risk-adjustment, we look at the difference in Sharpe ratios, i.e.

$$\Delta SR_{i,j} = SR(f^j_i) - SR(f_i). \tag{3}$$

Since some of our timing strategies utilize leverage, we note that Sharpe ratios are unaffected by leverage because both the numerator and the denominator are proportionally affected. Therefore, leverage does not mechanically enhance timing success. Statistical significance of the difference in Sharpe ratios can easily be assessed using the test of Jobson and Korkie (1981) of testing the null that $\Delta SR_{i,j} = 0$.

We also assess the performance of timed factors by calculating the time-series alpha by regressing the timed factor returns on the untimed ones (see, e.g., Gupta and Kelly, 2019):

$$f^j_{i,t+1} = \alpha_{i,j} + \beta_{i,j} f_{i,t+1} + \epsilon_{t+1}. \tag{4}$$

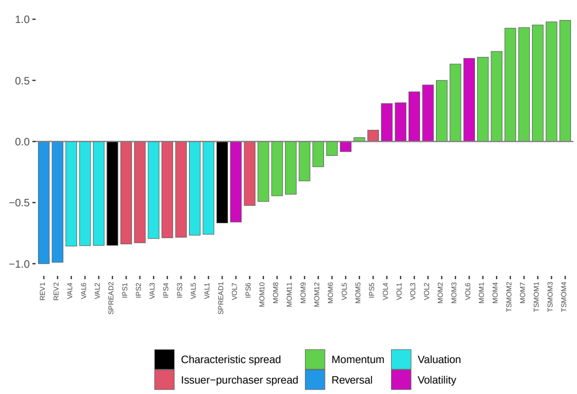
The alphas must be interpreted with caution, as they are in general affected by the leverage chosen for the timed strategy. However, the statistical significance of alphas is not influenced by leverage and implies that, for statistically significant alphas, the managed strategy expands the efficient frontier.

Figures 2 and 3 give a first overview of the univariate timing results. Figure 2 displays the net fraction of significant performance differences, obtained as the fraction of factors with

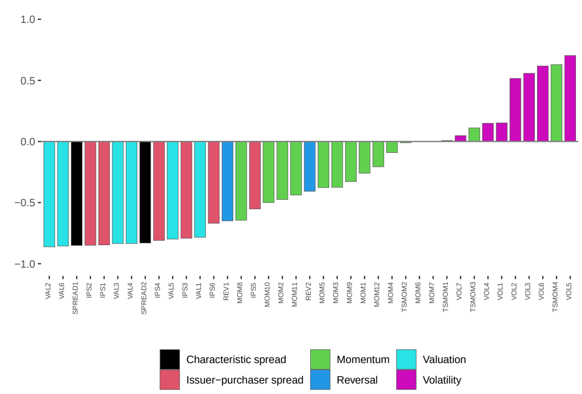
Figure 2: Timing Success by Signal Category (Returns and Sharpe Ratios)

This figure shows for each timing signal the fraction of factor portfolios with significant positive performance difference between the timed (f_i^j) and untimed (f_i) factors minus the corresponding fraction of significant negative performance differences. Colors indicate the timing category. Table B.1 provides a description of the individual timing signals and the assigned signal class. Figure (a) displays the net fraction for mean returns, Figure (b) for Sharpe ratios. We determine statistical significance at the 5 percent level. For Sharpe ratios, we use the z-statistic from the Jobson and Korkie (1981) test of the null that $\Delta SR_{i,j} = SR(f_i^j) - SR(f_i) = 0$.

(a) Timing effects on mean returns



(b) Timing effects on Sharpe Ratio



significant positive performance differences between the timed and the untimed portfolios minus the fraction of factors with significant negative performance differences. Panel (a) displays the measure for average returns. We find that timing signals based on momentum lead to the largest improvements. There are some exceptions, such as the signals based on Ehsani and Linnainmaa (2022), which by construction lead to a low average exposure to the original factor.[2] Panel (b) shows that for most signals factor timing on average decreases Sharpe ratios. Only volatility signals are able to improve Sharpe ratios. The top signals are based on the standard deviation of the previous month's daily returns (Moreira and Muir, 2017) and on S&P 500 implied volatility (Reschenhofer and Zechner, 2024). Time series momentum with 12 months lookback period (Moskowitz et al., 2012) also delivers strong performance. All other signals have weaker results.

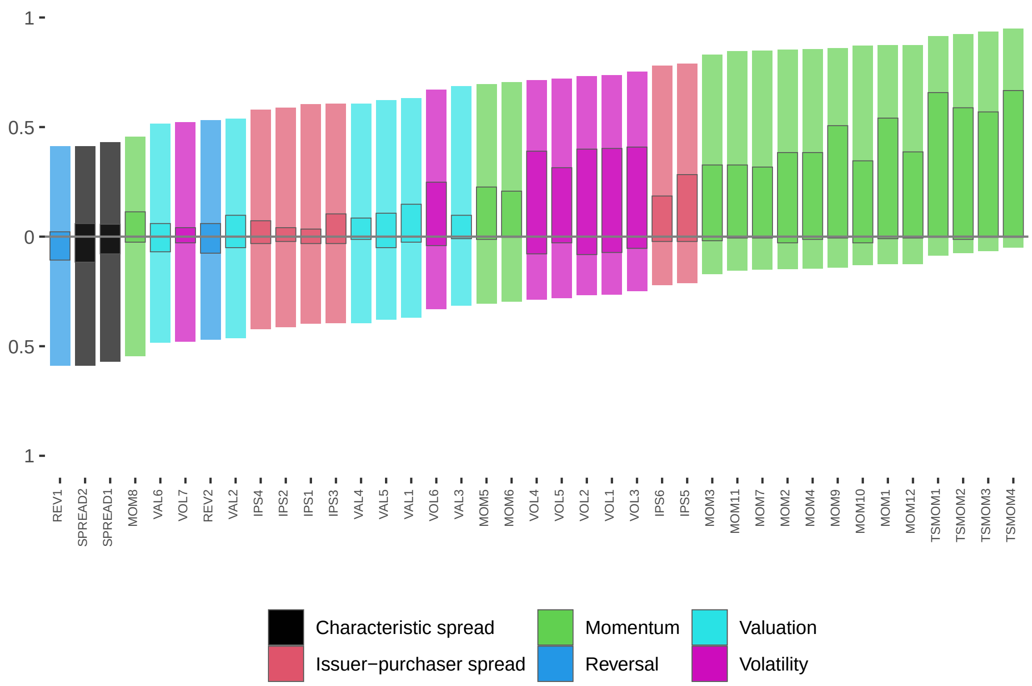
Figure 3 presents the univariate timing results, as estimated by the alphas in Eq. (4).

[2] These signals assign a weight of ± 1 in any given period, therefore the average exposure over time cannot exceed 1 and will typically be smaller than 1.

It plots the fraction of factor portfolios with positive and negative alphas, respectively, for each timing signal. Each bar has a length of 1; the vertical position of the bar shows the fraction of positive and negative alphas. Areas with dark borders within a bar present the fraction of timed factors with statistically significant α. We use a 5 percent significance level, with t-statistics adjusted for heteroscedasticity. The signals are ranked according to the fraction of positive alphas. Momentum signals achieve the highest fraction of positive alphas. More importantly, positive alphas tend to be statistically significant, while there is almost no statistical significance for negative alphas. The single best momentum signal is time series momentum with 12 months lookback period, as defined in Moskowitz et al. (2012). Volatility timing signals achieve performance improvements in the same ballpark as momentum, with high percentages of statistically significant positive alphas. The top signal in this group is the standard deviation of the previous month's daily returns, as described in Moreira and Muir (2017). Timing signals based on valuation, reversal, characteristics spreads and issuer-purchaser spread are less successful.

Figure 3: Timing Success by Signal Category (Alpha)

This figure presents the univariate timing results, measured by the alphas $\alpha_{i,j}$ from an OLS regression of timed factor portfolios' excess returns $f^j_{i,t+1}$ on unmanaged factor portfolio's excess returns $f_{i,t+1}$: $f^j_{i,t+1} = \alpha_{i,j} + \beta_{i,j} f_{i,t+1} + \epsilon_{t+1}$. For each timing signal, the distribution of alphas is shown with positive alphas above the horizontal zero-line and negative alphas below it. The bars have a total length of 1, with the vertical position indicating the relative magnitude of positive and negative alphas. Colors indicate the signal class. The dark shaded areas of the bars present the fraction of $\alpha_{i,j}$ significant at the 5 percent level, using t-statistics adjusted for heteroscedasticity. Table B.1 provides a description of the individual timing signals and the assigned signal class.



3.1.2 Timing Performance for Different Categories of Factors

In the previous analysis, we aggregated performance information across all 318 factors for different timing signals. While some level of aggregation is clearly necessary for tractability, it may mask important heterogeneity in timing success across factors. Factors that capture different sources of risk can potentially be timed with different signals. We therefore use the economic interpretation of factors to group them into seven categories: intangibles, investment, momentum, profitability, trading frictions, value vs. growth, and other.[3] We

[3] See Table A.1 for further details.

compile the results for categories of factors in Table 1. The panels show results for all signals (Panel A), momentum signals (Panel B), and volatility signals (Panel C).[4] We display average alphas of time-series regressions and differences in Sharpe ratios. We report simple averages over all factors within an economic category and for signals of a given type. Average t-statistics and counts of statistically significant factors in brackets are based on heteroscedasticity-adjusted standard errors.

The average annualized alpha across all factors and all signals equals 3.2%. This number is economically large, but there is rather weak statistical significance. The average t-statistic is just below 1. For the average signal, out of 318 factors, alphas for 84 are statistically significantly positive and for 11 are statistically significantly negative at the 5% level. There is strong heterogeneity across factor categories. Timing profitability factors produces the highest average alpha of 5.0%. This contrasts with an average alpha of 2.3% for factors related to investment and to trading frictions. Column ΔSR shows the average difference in the Sharpe ratio of the timed versus original factor. We show the average z-statistic from the Jobson and Korkie (1981) test of the null that $\Delta SR = 0$ in square brackets. The negative values indicate that a "brute force" application of all signals on all factors tends to reduce risk-adjusted returns.

Panel B reports results for timing using only momentum signals. This shows more successful factor timing. In particular, average alphas for profitability and value vs. growth factors are economically high (10% p.a.) and statistically significant; the corresponding change in Sharpe ratios is positive. The fraction of positive alphas is highest for investment, profitability, and value vs. growth factors. Timing momentum factors with momentum signals is less attractive: Average alphas are statistically insignificant, and the average change in Sharpe ratios is negative. With the exception of profitability factors, no economic category has more than 50% positive differences in the Sharpe ratio, out of which few are statistically significant. It seems that momentum signals enhance the performance, but increase volatility

[4] We relegate details for signals based on the the characteristic spread, issuer-purchaser spread, reversal and valuation to appendix Table C.1.

even more. Strategies based on those signals might be useful if they constitute only a small part of the portfolio.

Panel C shows that while timing with volatility signals leads to smaller gains in alphas, a higher proportion of alphas is statistically significant. Further, timing with volatility signals enhances Sharpe ratios. Volatility signals work best for momentum factors, where the average Sharpe ratio gain of 0.2 is statistically significant.

Table 1: Performance Impact of Factor Timing with Single Signals

This table shows timing success of different signals for individual factors, grouped into economic categories. N_f reports the number of factors within each category. The left part of the panel shows the alpha for each factor i and signal j against its raw (untimed) counterpart. Alpha is obtained as the intercept in the following regression: $f^j_{i,t+1} = \alpha_{i,j} + \beta_{i,j} f_{i,t+1} + \epsilon_{t+1}$. α, $\alpha > 0$, and $\alpha < 0$ present the average alpha, and the number of factors with a positive and negative α, respectively. We report average t-statistics and the number of significant factors at the 5 percent level in brackets, based on heteroscedasticity-adjusted standard errors. The right part shows the average difference in the annualized Sharpe ratio of the timed versus untimed factor across factor/signal combinations. For Sharpe ratios, we use the z-statistic from the Jobson and Korkie (1981) test of the null that $\Delta SR_{i,j} = SR(f^j_i) - SR(f_i) = 0$. Panel A displays the simple averages over all signals. Panels B and C report results for momentum and volatility signals. Table C.1 shows results for the remaining timing signal types. We describe the factors and their allocation to an economic category in Table A.1. Table B.1 describes the timing signals.

		Time series regression			Sharpe ratio difference		
	N_f	α	$\alpha > 0$	$\alpha < 0$	ΔSR	$\Delta SR > 0$	$\Delta SR < 0$
A. All signals							
All factors	318	3.190 [0.965]	224 [84]	94 [11]	-0.124 [-0.689]	114 [30]	204 [79]
Intangibles	53	2.731 [0.817]	36 [12]	17 [2]	-0.126 [-0.674]	20 [3]	33 [12]
Investment	46	2.250 [0.916]	33 [12]	13 [2]	-0.158 [-0.984]	14 [5]	32 [16]
Momentum	22	3.815 [1.282]	16 [7]	6 [0]	-0.316 [-1.510]	6 [3]	16 [9]
Profitability	35	4.952 [1.319]	27 [12]	8 [1]	-0.054 [-0.225]	16 [4]	19 [6]
Trading frictions	46	2.294 [0.660]	30 [10]	16 [2]	-0.096 [-0.505]	17 [5]	29 [10]
Value vs. growth	41	4.344 [1.268]	30 [15]	11 [2]	-0.091 [-0.592]	15 [4]	26 [9]
Other	75	3.004 [0.865]	52 [16]	23 [2]	-0.111 [-0.661]	26 [5]	49 [17]
B. Momentum signals							
All factors	318	6.428 [1.615]	264 [130]	54 [4]	0.001 [-0.220]	126 [34]	192 [52]
Intangibles	53	5.843 [1.558]	44 [21]	9 [0]	0.008 [-0.134]	24 [5]	29 [7]
Investment	46	4.951 [1.773]	42 [20]	4 [0]	0.018 [-0.273]	15 [6]	31 [10]
Momentum	22	5.413 [1.153]	17 [6]	5 [0]	-0.164 [-1.233]	5 [1]	17 [7]
Profitability	35	9.937 [2.061]	32 [19]	3 [0]	0.078 [0.292]	19 [5]	16 [3]
Trading frictions	46	3.865 [0.837]	30 [12]	16 [2]	-0.052 [-0.499]	16 [5]	30 [11]
Value vs. growth	41	9.939 [2.370]	37 [24]	4 [0]	0.067 [0.139]	17 [4]	24 [4]
Other	75	6.059 [1.549]	62 [27]	13 [0]	-0.005 [-0.214]	29 [8]	46 [10]
C. Volatility signals							
All factors	318	1.300 [1.079]	220 [100]	98 [17]	0.026 [0.383]	183 [57]	135 [24]
Intangibles	53	0.464 [0.587]	35 [12]	18 [4]	0.000 [0.003]	28 [5]	25 [5]
Investment	46	0.507 [0.854]	31 [12]	15 [3]	-0.002 [0.059]	25 [5]	21 [5]
Momentum	22	5.035 [3.042]	20 [16]	2 [0]	0.154 [2.053]	19 [12]	3 [0]
Profitability	35	2.512 [1.597]	28 [12]	7 [1]	0.085 [1.022]	24 [9]	11 [1]
Trading frictions	46	0.910 [0.894]	30 [15]	16 [5]	0.017 [0.270]	25 [10]	21 [5]
Value vs. growth	41	1.675 [1.719]	34 [19]	7 [2]	0.060 [0.833]	30 [9]	11 [1]
Other	75	0.750 [0.509]	42 [14]	33 [4]	-0.014 [-0.114]	33 [7]	42 [6]

3.2 One Factor - Many Signals

Section 3.1 suggests heterogeneity in timing success: The extent to which factor timing works appears to be factor and signal specific. Clearly we cannot feasibly analyze the combination of 318 factors $\times$ 39 signals in a simple manner but need to resort to dimension reduction techniques. In a first step of aggregation, we still time each factor individually, but we use multiple signals to make a timing decision. Since many of the signals are highly correlated, it is clear that we should not simply run a "kitchen sink" regression and expect to obtain sensible predictions. We therefore apply partial least squares (PLS) as the appropriate signal aggregation technique. We briefly introduce PLS in the next section and refer to Kelly and Pruitt (2013, 2015) for a comprehensive treatment.

3.2.1 Partial Least Squares

For the aggregation of the right-hand side, we could use principal components analysis (PCA), a well-known statistical approach that is widely applied in finance. Intuitively, PCA extracts $k < J$ linear combinations of the original $J = 39$ signals in a way to explain as much as possible of the variance of the original signals. Yet, our goal is not primarily a parsimonious description of the signals per se, but to find an efficient set of predictors for time-varying factor returns. Hence, we resort to a related technique that is better suited to be used in a predictive regression setting – partial least squares. Kelly and Pruitt (2013) use PLS to successfully predict the market index.[5] The main idea of PLS in our setting is to find linear combinations of the original signals that maximize their covariances with the factor return. More precisely, consider the regression model

$$f_i = W_i \beta_i + \epsilon_i \, , \tag{5}$$

where f_i is a $T \times 1$ vector of factor i's one-period ahead excess returns, and T is the sample

[5] Light, Maslov, and Rytchkov (2017) employ PLS successfully for cross-sectional predictions.

length. W_i is a $T \times J$ factor-specific signal matrix that contains $J = 39$ column vectors w_i^j, β_i is a $J \times 1$ vector of signal sensitivities and ϵ_i is a $T \times 1$ vector of errors. PLS decomposes W_i such that the first k vectors can be used to predict f_i. We can write this as

$$f_i = \left(W_i P_i^k\right) b_i^k + u_i \,. \tag{6}$$

P_i^k is a $J \times k$ matrix with columns $v_m,\ m = 1, \ldots, k$, and b_i^k is a $k \times 1$ vector of sensitivities to the aggregated signals. To find the vectors v_m, we iteratively solve the following problem

$$v_m = \arg\max_v [cov(f_i, W_i v)]^2, \ \text{ s.t. } v'v = 1, \ \ cov(W_i v, W_i v_n) = 0 \ \forall \ n = 1, 2, ..., m-1.^{6} \tag{7}$$

PLS is well suited for problems such as factor timing as it can deal with highly correlated signals. In particular, a linear combination of the signals can be identified as a useful predictor of factor returns even if it does not explain much of the variation among signals.

3.2.2 Univariate Factor Timing with PLS

We produce one-month ahead forecasts using standard predictive regressions of the dominant components of factor returns. For each one of 314 factors,[7] we run four PLS regressions as specified in Eq. (6), where the number of components k equals 1, 2, 3, and 5. We use each factor's first half of the sample to obtain initial estimates, and use the second half to form out-of-sample (OOS) forecasts to ensure that our OOS results are not subject to look-ahead bias. As in Campbell and Thompson (2008), we use monthly holding periods and calculate out-of-sample R^2 as

[6] Note that we run a separate PLS regression for each factor to capture differential dynamics in factor risk premia. To emphasize this procedure, we could write $v_m^{(i)}$ to emphasize the dependence on i. In order to ease the notation, we omit this superscript.

[7] We lose 4 factors due to lack of sufficient historical data. These are: Activism1, Activism2, Governance, and ProbInformedTrading.

$$R^2_{OOS} = 1 - \frac{\sum_{t=1}^{T}\left(f_{i,t+1} - \hat{f}_{i,t+1}\right)^2}{\sum_{t=1}^{T}\left(f_{i,t+1} - \bar{f}_{i,t+1}\right)^2}, \tag{8}$$

where $\hat{f}_{i,t+1}$ is the predicted value from a predictive regression estimated through period t, and $\bar{f}_{i,t+1}$ is the historical average return estimated through period t. To assess the economic importance of factor timing, we follow Campbell and Thompson (2008) and compare the average excess return that a buy-and-hold investor will earn from investing in factors without timing, $R = \frac{SR^2}{\gamma}$, where γ denotes the coefficient of relative risk aversion, to the average excess returns earned by an active investor exploiting predictive information through PLS regressions, obtained from

$$R^* = \frac{1}{\gamma}\frac{SR^2 + R^2_{OOS}}{1 - R^2_{OOS}}. \tag{9}$$

We follow Campbell and Thompson (2008) and assume unit risk aversion, i.e. $\gamma = 1$. Table 2 presents statistical and economic measures of timing success in the PLS framework. Panel A reports the average R^2_{OOS} of these regressions and the 25th, 50th, and 75th percentiles. Panel B groups the factors into economic categories and reports the average R^2_{OOS} per category. Panels C and D report average excess returns for all factors and economic categories, respectively. Panel A of Table 2 shows that the average out-of-sample R^2_{OOS} (over all factor portfolios) for partial least squares predictive regressions using just one component (PLS1) equals 0.76%, on a one-month prediction horizon. The median is almost identical to the mean. The average out-of-sample R^2_{OOS} turns negative when using more than a single component. This shows that factor timing is prone to overfitting. Panel B provides the results for different economic factor categories. For PLS1, all out-of-sample R^2_{OOS} are positive, ranging from barely positive (0.02 percent) for momentum factors to 1.4 and 1.6 percent for profitability and value vs. growth factors, respectively. Panels C and D show the corresponding returns for active investors with unit risk aversion who optimally exploit predictive information. The average excess return across all untimed factors equals 2.4%; this increases to 3.2% using PLS1. The increase in excess returns is pervasive but heterogeneous among economic categories. The largest gains are obtained for the profitability and value vs. growth factor categories. The gain for momentum-based factors is relatively meager. This heterogeneity in timing success results in timed factors corresponding to the value vs. growth category outperforming momentum factors, thus reversing the attractiveness of the untimed factors.

Table 2: Predictive Regressions of Factor Excess Returns

This table shows out-of-sample R^2_{OOS} and active investor excess returns obtained from predictive regressions of factor returns on timing signals. For each one of 314 factors, we run four partial least squares (PLS) regressions, where the number of components equals 1, 2, 3, and 5. Panel A reports the average R^2_{OOS} of these regressions and the 25, 50, and 75 percentiles. Panel B groups the factors into economic categories and reports the average R^2_{OOS} per category. Panel C compares the annualized average excess return R (R^*) that investors will earn when optimally choosing their factor exposure, either without timing or exploiting predictive information through PLS regressions. We follow Campbell and Thompson (2008) to determine untimed returns $R = SR^2/\gamma$, shown in column Untimed factors, and timed returns $R^* = (SR^2 + R^2_{OOS})/(\gamma(1 - R^2_{OOS}))$, shown in columns PLS 1 to PLS 5, assuming unit relative risk aversion γ. Panel D displays average active investor returns per economic factor category. We use the first half of the sample to obtain initial estimates, and report only values from out-of-sample regressions using an expanding window. We describe the factors and their allocation to an economic category in Table A.1. Table B.1 describes the timing signals.

	Untimed factors	PLS - timed factors			
N of components		1	2	3	5
A. Full sample R^2_{OOS}					
Mean		0.754	-0.218	-1.044	-2.058
25 perc.		-0.166	-1.186	-2.116	-3.133
50 perc.		0.757	0.097	-0.444	-1.290
75 perc.		1.793	1.285	0.886	0.352
B. Economic category R^2_{OOS}					
Intangibles		0.467	-0.809	-1.777	-2.447
Investment		0.789	-0.175	-0.572	-1.365
Momentum		0.017	-1.118	-1.401	-1.420
Profitability		1.404	0.283	-1.397	-3.781
Trading frictions		0.451	-0.838	-1.562	-2.761
Value vs. growth		1.612	1.185	0.456	-0.527
Other		0.551	-0.200	-1.064	-2.009
C. Full sample R (R^*)					
Mean	2.364	3.202	2.238	1.461	0.606
D. Economic category R (R^*)					
Intangibles	1.345	1.887	0.631	-0.274	-0.871
Investment	3.716	4.580	3.581	3.219	2.476
Momentum	4.706	4.773	3.724	3.440	3.437
Profitability	1.626	3.133	1.978	0.408	-1.228
Trading frictions	1.538	2.039	0.748	0.149	-0.781
Value vs. growth	3.151	4.908	4.503	3.773	2.813
Other	1.951	2.589	1.859	1.010	0.122

In practice, risk constraints or other frictions might prohibit an investor from fully exploiting the information of the signals. The results presented in Table 2 may thus appear as an overstatement. To alleviate this concern we construct the simplest possible univariate timing strategies. For each of the 314 factors, we define the timed portfolio return $f^{\tau}_{i,t+1}$ as follows:[8]

$$f^{\tau}_{i,t+1} = \begin{cases} +f_{i,t+1} & \text{if } \hat{f}_{i,t+1} \geq 0, \\ -f_{i,t+1} & \text{if } \hat{f}_{i,t+1} < 0. \end{cases} \tag{10}$$

The timed portfolio return is thus equal to the original factor return when the forecast is positive. In the event of a negative predicted return, the timed factor return is its negative value, i.e. the short-long untimed portfolio return. Return predictions are made using PLS regressions, where we again vary the number of components. In order to compute performance statistics, we use a two-step procedure: First, we compute statistics for each individual factor separately for its out-of-sample period. Second, we take cross-sectional averages. In this thought experiment, we do not take the perspective of an investor diversified across factors, but an investor who is randomly sampling one factor from the set of 314 factors.

Table 3 reports performance statistics for untimed factors and univariate factor timing portfolios. In Panel A we report the average return (R), standard deviation (SD), Sharpe ratio (SR), and maximum drawdown (maxDD). Timed portfolios using PLS1 exhibit significant economic gains to investors when compared to untimed portfolios: the average annualized performance increases from 4.0% to 5.9% p.a., the average Sharpe ratio increases from 0.34 to 0.48, and the average maximum drawdown decreases from 46% to 38%. Increasing the number of PLS-factors k does not provide additional benefits; all performance measures slowly worsen when increasing k. Panel B shows regression results, where we regress returns of the timed portfolio on the market excess return (CAPM) and the six factors from the Fama-French-5+Momentum asset pricing models. The CAPM-alpha increases from an average

[8]For ease of notation, we suppress the number of PLS components k.

4.5% for untimed factors to 6.6% for timed factor portfolios using PLS1. The corresponding multifactor alphas are 3.7% and 4.9%, respectively. Using the CAPM, the highest alphas are obtained for PLS1, while the highest multifactor alpha of 5.0% is obtained for PLS2. Panel C displays the timing success from time-series regression of the timed on untimed factor returns, a comparison of Sharpe ratios, and the hit rate (fraction of months where the sign of the predicted return corresponds to the sign of the realized return). Also using these measures, the best performance is achieved for timed portfolios using PLS1 (based on the return difference, increase in Sharpe ratio, and hit rate) and PLS2 (based on the alpha from timing regressions). The average hit rate using PLS1 equals 57%. Panel C further illustrates that prediction success is higher for positive factor returns than for negative ones. The hit rate conditional on positive realized returns is almost 80%, while the sign is forecast correctly only for 30% of negative realized returns.

Table 3: Univariate Factor Timing

This table shows results for univariate factor timing portfolios. For each factor i, the predicted returns ($\hat{f}_{i,t+1}$) are from partial least squares (PLS) regressions, where the number of components equals 1, 2, 3, or 5. The timed portfolio, denoted by $f^{\tau}_{i,t+1}$, is equal to the original factor return, when the forecast is positive. In the event of a negative predicted return, the timed factor return is essentially its inverse, i.e. the short-long untimed portfolio return. All statistics are obtained in a two-step procedure: First, we compute statistics for each individual factor separately for its out-of-sample period. Second, we take cross-sectional averages. Panel A reports the annualized average return in percent, annualized standard deviation in percent, annualized Sharpe ratio, and maximum drawdown in percent. Panel B shows regression results, where we regress returns on the CAPM and FF5+Momentum asset pricing models. We report average t-statistics in parenthesis. Panel C displays the timing success. α, ΔR, and ΔSR denote the time-series regression alpha, difference in return and difference in Sharpe ratio of the timed vs. untimed factor returns. We again report t- and z-statistics in parenthesis. % $\hat{f}_{i,t+1} > 0$ (% $\hat{f}_{i,t+1} < 0$) shows the fraction of positive (negative) predictions. $\hat{f}_{i,t+1} > 0 \mid f_{i,t+1} > 0$ ($\hat{f}_{i,t+1} < 0 \mid f_{i,t+1} < 0$) reports the fraction of positive (negative) return prediction conditional on the untimed factor returns' $f_{i,t+1}$ realisation being positive (negative). hit rate reports the fraction where the sign of the predicted return $\hat{f}_{i,t+1}$ corresponds to the sign of $f_{i,t+1}$. We describe the factors and their allocation to an economic category in Table A.1. Table B.1 describes the timing signals.

	Untimed factors	PLS - timed factors			
N of components		1	2	3	5
A. Performance					
R	4.040	5.879	5.665	5.460	5.003
SD	12.782	12.752	12.751	12.767	12.78
SR	0.336	0.482	0.469	0.444	0.412
maxDD	45.963	37.881	38.548	39.177	40.267
B. Regression results					
CAPM α	4.515	6.588	6.266	6.008	5.52
	(2.179)	(3.042)	(2.938)	(2.765)	(2.559)
FF5 + MOM α	3.368	4.938	4.990	4.833	4.503
	(1.950)	(2.497)	(2.439)	(2.277)	(2.126)
C. Timing success					
α		2.354	2.517	2.440	2.148
		(1.112)	(1.169)	(1.077)	(0.994)
ΔR		1.840	1.625	1.420	0.963
		(0.490)	(0.382)	(0.255)	(0.119)
ΔSR		0.147	0.133	0.109	0.076
		(0.481)	(0.380)	(0.250)	(0.114)
% $\hat{f}_{i,t+1} > 0$		74.366	71.196	70.485	69.031
% $\hat{f}_{i,t+1} < 0$		25.634	28.804	29.515	30.969
$\hat{f}_{i,t+1} > 0 \mid f_{i,t+1} > 0$		79.025	75.966	74.870	73.424
$\hat{f}_{i,t+1} < 0 \mid f_{i,t+1} < 0$		29.905	33.386	33.699	34.764
hit rate		56.930	56.813	56.350	56.034

In order to better understand the heterogeneity in timing success, we illustrate the timing of selected individual factors in Table 4. Specifically, we highlight the best and worst univariate factor timing results. Panel A selects factors conditional on the sign of average returns of the original factor and sorts them on the difference between timed and untimed factor returns. We find that for some factors, a (considerable) negative unconditional return can be transformed into substantial positive timed return. In contrast, as highlighted in Panel B, it is rarely the case that a positive unconditional risk premium turns negative through poor timing decisions. While sometimes returns are indeed negative, only one factor out of 314 has significantly negative return differences (OrderBacklogChg).

Table 4: Best and Worst Univariate Timing Results (Using PLS1)

This table shows factors with the best and worst univariate timing results. We report average untimed (f_i) and timed factor returns (f_i^τ), as well as their return differences (ΔR), and t-statistics ($t(\Delta\text{R})$ of these differences, respectively. Panel A selects the top 10 factors where the average untimed return is negative, conditional on the timed factor return being positive, and sorts results based on the t-statistic of the difference. Panel B displays all factors where the untimed return is positive, conditional on the timed factor return being negative. Results are again sorted by the t-statistic of return differences. We describe the factors in Table A.1.

	f_i	f_i^τ	ΔR	$t(\Delta\text{R})$
A. $f_i < 0 \mid f_i^\tau > 0$				
EntMult_q	-12.265	14.128	26.394	6.562
ChNCOA	-9.041	8.948	17.989	6.172
sgr	-6.867	5.649	12.516	4.828
ChNCOL	-6.077	5.632	11.710	4.714
AssetGrowth_q	-9.272	11.442	20.713	4.479
EarningsPredictability	-7.407	10.120	17.527	4.346
NetDebtPrice_q	-14.899	13.562	28.461	4.160
ReturnSkewCAPM	-3.043	2.998	6.041	3.607
betaRC	-2.655	6.427	9.082	3.072
AbnormalAccrualsPercent	-2.118	1.559	3.677	2.469
...				
B. $f_i > 0 \mid f_i^\tau < 0$				
OrderBacklogChg	4.128	-2.256	-6.384	-2.984
BrandInvest	3.259	-1.037	-4.296	-1.038
ChNAnalyst	22.191	-5.647	-27.838	-1.012
EBM	0.806	-0.398	-1.205	-0.893
DelayNonAcct	1.467	-0.538	-2.005	-0.844
PctTotAcc	0.291	-0.322	-0.613	-0.754

3.3 Multifactor Timing

The previous analyses show that factor timing can be beneficial even if applied only to individual factors. However it is unlikely that a sophisticated investor seeks exposure to only one source of systematic risk. We therefore investigate the gains of factor timing for an investor who seeks exposure to multiple factors. Further, the univariate timing strategy displayed in Table 3 makes only use of the sign of the prediction, while investors can benefit from strategies that are conditional on the sign and magnitude of the predicted factor returns.

One easy way to consider magnitudes is to sort factors into portfolios based on the predicted returns. Below we will consider quintile portfolios. Each month t we sort factors into five portfolios based on their $t+1$ predicted excess return from partial least squares regressions with 1 component. In addition, we construct a high-low (H-L) portfolio. To compare the performance of the timed quintile portfolios, we construct quintile portfolios which are sorted based on the historical average factor return. We name these benchmark portfolios 'naively timed'. It is plausible that investors expect factors that have historically performed well (poorly), to do so again in the future. This implementation also directly addresses the concern that factor timing might only be successful because we are capturing factors with high unconditional returns. Thus, the benchmark portfolios are formed as

$$f_{t+1}^{u_q} = w_{i,t}^{u_q} f_{i,t+1} \tag{11}$$

where $w_{i,t}^{u_q}$ equals $1/N_t^q$ for factors where the historical average return up to t is in the qth quintile, and 0 otherwise. The quintile portfolios of timed factors are given by

$$f_{t+1}^{PLS1_q} = w_{i,t}^{PLS1_q} f_{i,t+1}\,, \tag{12}$$

where f^{PLS1_q} is the quintile q PLS1 portfolio and the weight $w_{i,t}^{PLS1_q}$ equals $1/N_t^q$ for all factors where the $t+1$ return predicted with $PLS1$ is in the qth quintile, and 0 otherwise.

Table 5 displays the results. Panel A reports performance statistics of the quintile and the high-low (H-L) portfolios, both for the naively timed and the PLS1-timed sorts. We find several interesting results. First, sorting factors into portfolios merely based on their historical average return leads to a monotonic increase in performance across sorted portfolios. In other words, past factor returns seem to be a good predictor for future factor returns. The High (Low) portfolio, for example, produces an annualized average return of 10.50 (1.16) percent. Hence, portfolio sorts based on the historical average constitute a tough benchmark. The H-L portfolio delivers an average return of 9.34% p.a. Second, timing improves returns and Sharpe ratios, and helps reduce drawdowns. The predicted returns for the timed portfolios range from -5.77% to 16.91%. Realized returns spread across a slightly narrower interval from -3.92% to +14.07%, which is more pronounced than the return spread in the untimed portfolios. The top PLS1-based quintile leads to the highest Sharpe ratio of about 1.81, compared to 1.49 for the top portfolio based on historical averages. Panel B displays CAPM and FF5+Momentum alphas. Alphas of untimed factor portfolios increase monotonically from 0.95% to 11.39%. For timed portfolios, the increase is much more pronounced and ranges from -4.38% for the lowest quintile to +14.92% for the highest quintile. Results are similar for FF5+Momentum alphas, although the magnitudes are slightly lower. Panel C shows that alphas from regressing the returns of the timed quintile portfolios on the original factor returns increase monotonically. The return difference between the PLS1-timed H-L portfolio and its naively timed counterpart amounts to 8.65%. Thus, the timed H-L portfolio earns twice the return of the comparable naively timed portfolio. While less pronounced, the difference in Sharpe ratios equals still impressive, 0.44, and it is statistically significant. The reason for the more moderate difference in Sharpe ratios (1.23 timed, 0.79 untimed) is the higher standard deviation of the timed H-L portfolio returns.

Table 5: Portfolio Sorts Based on Predicted Factor Returns

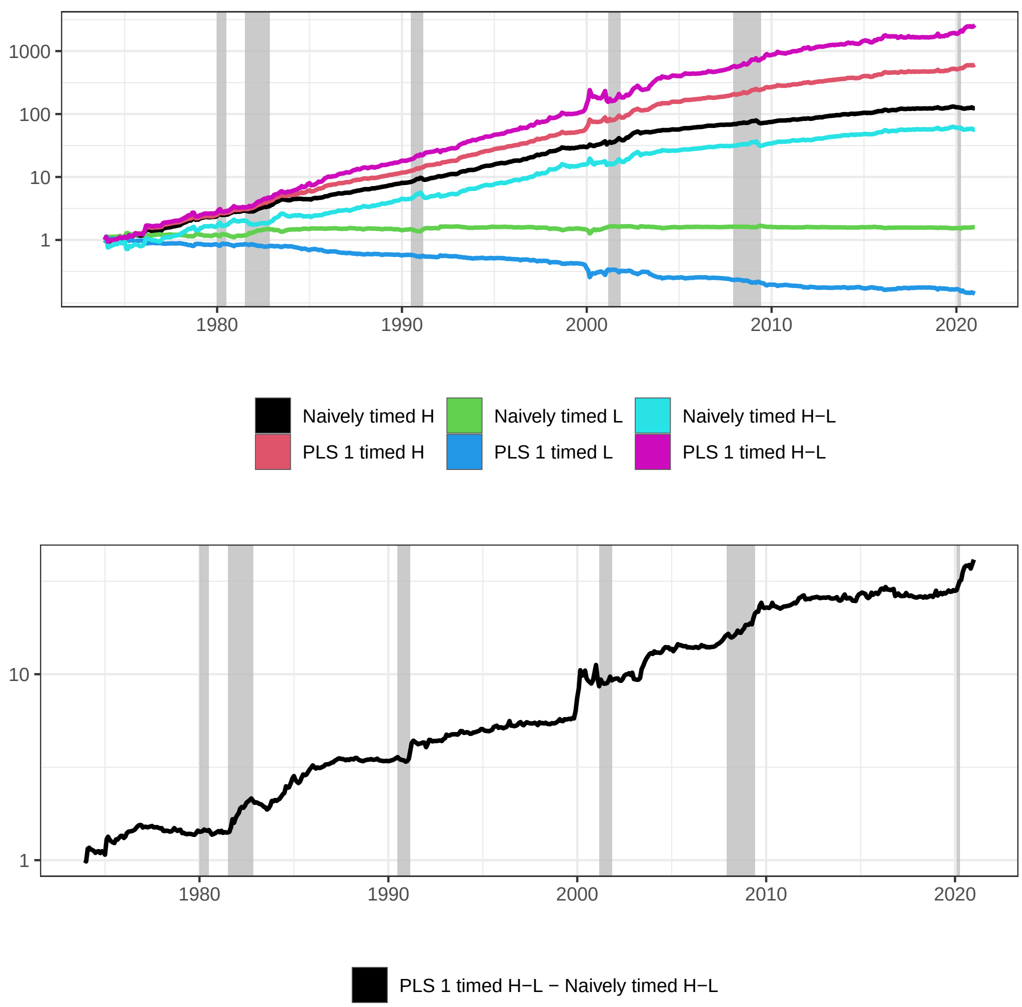
This table shows performance statistics for sorted portfolios based on the predicted factor returns. At the end of each month t, we sort factors into 5 portfolios based on their $t+1$ predicted return and construct a high-low (H-L) portfolio. For naively timed factor sorts, portfolios are constructed based on the historical average return. PLS1 timed factor sorts are based on predictions from partial least squares regressions with 1 component. In Panel A, we report the annualized mean predicted return (Pred), realized return (R), standard deviation (SD), Sharpe ratio (SR) and maximum drawdown (maxDD). Panel B shows regression results, where we regress returns on the CAPM and FF5+Momentum asset pricing models. We report average t-statistics in brackets. Panel C displays the timing success. α, ΔR, and ΔSR exhibit the time-series regression alpha, difference in return and difference in Sharpe ratio of the timed vs. untimed factor returns. We again report t- and z-statistics in brackets. We describe the factors and their allocation to an economic category in Table A.1. Table B.1 describes the timing signals.

	Naively timed portfolio sorts						PLS1 timed portfolio sorts					
	L	2	3	4	H	H-L	L	2	3	4	H	H-L
A. Portfolio performance												
Pred							-5.771	1.129	4.599	8.263	16.907	22.678
R	1.161	3.651	4.112	5.459	10.501	9.341	-3.918	2.175	5.399	7.333	14.072	17.99
SD	6.18	3.397	2.997	5.03	7.058	11.888	7.658	3.314	3.427	4.939	7.797	14.678
SR	0.188	1.075	1.372	1.085	1.488	0.786	-0.512	0.656	1.575	1.485	1.805	1.226
maxDD	22.943	7.176	5.979	13.166	15.278	37.287	86.685	14.578	6.332	10.794	13.29	35.242
B. Regression results												
CAPM α	0.950	3.669	4.550	6.427	11.391	10.441	-4.357	2.493	6.116	7.983	14.923	19.281
	(1.048)	(7.343)	(10.850)	(9.507)	(11.382)	(6.089)	(-3.897)	(5.222)	(13.528)	(11.438)	(13.372)	(9.088)
FF5 + MOM α	1.660	3.632	3.559	3.464	7.232	5.572	-2.521	1.688	4.508	4.876	11.181	13.701
	(1.948)	(7.405)	(8.936)	(6.062)	(8.907)	(3.688)	(-2.456)	(3.629)	(11.141)	(8.469)	(11.864)	(7.382)
C. Timing success												
α							-4.803	1.545	2.900	3.646	5.801	10.238
							(-5.449)	(3.105)	(6.372)	(6.657)	(6.689)	(6.303)
ΔR							-5.079	-1.476	1.287	1.874	3.571	8.649
							(-5.612)	(-2.354)	(2.822)	(3.267)	(4.325)	(5.373)
ΔSR							-0.699	-0.418	0.203	0.399	0.317	0.440
							(-5.420)	(-2.197)	(1.352)	(3.273)	(2.624)	(3.591)

We further find that our timing approach offers robust performance over time. Figure 4 displays the performance for sorting factors on past average returns and factor timing portfolio sorts. The performance of our timing model using one component (i.e. PLS1) clearly and consistently outperforms portfolios sorted on historical average returns. For the High portfolio, the end-of-period wealth is about ten times larger than the comparable portfolio based on historical averages. Furthermore, the figure illustrates that the lowest quintile experiences negative returns on average. McLean and Pontiff (2016) find that many anomalies have lower average returns post-publication. And indeed, we find that the performance for "naively timed" gets flatter after the year 2000, i.e., sorting on the historical mean produces a smaller performance spread. Yet sorting on returns predicted from timing signals continues to work at least as well in recent periods as before 2000. We will discuss post-publication effects in more detail in Section 3.4.

Figure 4: Sorted Factor Portfolios - Cumulative Excess Returns

This figure displays the performance for factor timing portfolio sorts. We sort factors into quintile portfolios based on their $t+1$ predicted return and plot performance of the H (high), L (low) and H-L (high-low) portfolios. The total return indices are in excess of the risk-free rate. 'Naively timed' displays results for portfolio sorts based on the historic average. 'PLS 1 timed' shows results based on predictions from partial least squares regressions with 1 component.



To illustrate the dynamics of factor allocation, Table 6 shows the frequency of the 10 factor portfolios most held in the top and bottom portfolios. Panel A shows frequencies for naively timed factors, where factors are sorted into a quintile portfolio based on the historical average. We find a persistent presence of factors. ReturnSkewCAPM and betaCR, for example, are in the low quintile more than 9 out of 10 times; STreversal, MomSeasonShort, IntMom, MomOffSeason and IndRetBig end up in the top quintile about 90% of the time. Timing results in a more heterogeneous allocation. Even though ReturnSkewCAPM remains the most frequently represented factor in the low quintile, its presence is reduced by 11 percentage points. BetaCR drops to 56 percent, and is replaced by sgr as the second-most frequent factor. We find similar results for the high quintile: Compared to the naively timed portfolio, the top frequencies of the most held factors are lower, illustrating more pronounced dynamics in portfolio sorts.

Table 6: Top 10 Factor Investments

This table shows the frequency of factor allocation into the quintile portfolios. At the end of each month t, we sort factors into 5 portfolios based on their $t+1$ predicted return. Panel A shows frequencies for naively timed factors, where portfolios are sorted based on the historic average. Panel B shows results for PLS1 timed factor sorts, where predictions are based on partial least squares regressions with 1 component. We report the 10 factors with the lowest (L %) and highest (H %) frequencies in the Low and High quintile portfolio, respectively. $\Delta\%$ reports the difference of PLS1 timed allocations compared to allocation based on the historical average. We describe the factors and their grouping into an economic category in Table A.1.

Acronym	L %	Δ%	Acronym	H %	Δ%
A. Naively timed					
ReturnSkewCAPM	99		STreversal	100	
betaCR	92		MomSeasonShort	98	
BetaDimson	84		IntMom	97	
BetaFP	82		MomOffSeason	97	
BetaSquared	81		IndRetBig	94	
betaRR	79		ResidualMomentum	85	
IdioVolCAPM	78		MomVol	76	
ChNCOA	73		Mom12mOffSeason	74	
ChNCOL	73		EntMult	72	
sgr	73		AssetGrowth	71	
B. PLS1 timed					
ReturnSkewCAPM	88	-11	IndRetBig	95	1
sgr	71	-2	STreversal	90	-10
ChNCOA	70	-3	IntMom	76	-21
ChNCOL	67	-5	ResidualMomentum	62	-23
BetaDimson	57	-27	MomVol	58	-17
betaCR	56	-36	MomSeasonShort	58	-39
BetaSquared	53	-28	MomOffSeason	58	-39
BetaFP	53	-29	Mom12m	56	-13
FirmAge	52	2	Frontier	55	-6
VarCF	50	-15	MomRev	55	-3

3.4 Pre- vs. Post Publication Analysis

It is well documented that many factors have yielded lower average returns post their publication (McLean and Pontiff, 2016). A related concern is raised in Li, Rossi, Yan, and Zheng (2022). They document that machine learning predictability is considerably lower in real time, i.e. if signals are used only after they have been published. Table A.1 in the Appendix documents the publication date for each signal that is used for factor construction. To explore possible pre- and post publication effects for factor timing, we therefore repeat the analysis, but at every point in time, we use only factors that were already known, i.e. whose underlying paper had been published.

We first review the univariate timing results pre and post publication. Table 7 shows the univariate timing results for the full sample and also separately for the pre- and post publication period. The pre- vs. post comparison of the average factor returns clearly show that the average factor return is 3 percentage points lower than in the pre publication period (5.51% vs. 2.47%). However, the difference between the untimed and timed average factor remains almost constant in the pre and post publication period. In both samples, the timed factor returns are on average about 1.5 percentage points larger than their untimed counterparts. This suggests that even though the factor premia have decreased in levels, their time-variation still seems to follow similar dynamics that can be partially predicted.

Table 7: Univariate Factor Timing pre- and post-publication

This table shows pre- and post-publication results for univariate factor timing portfolios. We split the sample period based on each factor's individual publication date. For each factor i, the predicted returns ($\hat{f}_{i,t+1}$) are from partial least squares (PLS) regressions, where the number of components equals 1. The timed portfolio, denoted by $f^{\tau}_{i,t+1}$, is equal to the original factor return, when the forecast is positive. In the event of a negative predicted return, the timed factor return is essentially its inverse, i.e. the short – long untimed portfolio return. All statistics are obtained in a two-step procedure: First, we compute statistics for each individual factor separately for its out-of-sample period. Second, we take cross-sectional averages. Panel A reports the annualized average return in percent, annualized standard deviation in percent, annualized Sharpe ratio, and maximum drawdown in percent. Panel B shows regression results, where we regress returns on the CAPM and FF5+Momentum asset pricing models. We report average t-statistics in parenthesis. Panel C displays the timing success. α, ΔR, and ΔSR denote the time-series regression alpha, difference in return and difference in Sharpe ratio of the timed vs. untimed factor returns. We report t- and z-statistics in parenthesis. % $\hat{f}_{i,t+1} > 0$ (% $\hat{f}_{i,t+1} < 0$) shows the fraction of positive (negative) predictions. $\hat{f}_{i,t+1} > 0 \mid f_{i,t+1} > 0$ ($\hat{f}_{i,t+1} < 0 \mid f_{i,t+1} < 0$) reports the fraction of positive (negative) return conditional on the factor returns' $f_{i,t+1}$ realisation being positive (negative). hit rate reports the fraction where the sign of the predicted return $\hat{f}_{i,t+1}$ corresponds to the sign of $f_{i,t+1}$. We describe the factors, their publication date and their allocation to an economic category in Table A.1. Table B.1 describes the timing signals.

	Full-sample		Pre-publication		Post-publication	
	Untimed	PLS1 timed	Untimed	PLS1 timed	Untimed	PLS1 timed
A. Performance						
R	4.04	5.879	5.516	7.338	2.468	4.045
SD	12.782	12.752	12.619	12.591	11.984	11.952
SR	0.336	0.482	0.514	0.634	0.202	0.333
maxDD	45.963	37.881	33.119	27.849	38.118	31.603
B. Regression results						
CAPM α	4.515	6.588	5.953	7.895	2.916	4.889
	(2.179)	(3.042)	(2.237)	(2.737)	(0.966)	(1.632)
FF5 + MOM α	3.368	4.938	4.566	5.63	2.53	4.035
	(1.950)	(2.497)	(1.886)	(2.111)	(1.000)	(1.432)
C. Timing success						
α		2.354		2.116		1.818
		(1.112)		(0.615)		(0.656)
ΔR		1.840		1.822		1.576
		(0.490)		(0.265)		(0.334)
ΔSR		0.147		0.120		0.131
		(0.481)		(0.239)		(0.325)
% $\hat{f}_{i,t+1} > 0$		74.336		77.790		73.187
% $\hat{f}_{i,t+1} < 0$		25.634		22.210		26.813
$\hat{f}_{i,t+1} > 0 \mid f_{i,t+1} > 0$		79.025		82.454		75.619
$\hat{f}_{i,t+1} < 0 \mid f_{i,t+1} < 0$		29.905		28.311		31.268
hit rate		56.930		59.269		54.788

In the next analysis we repeat the multifactor timing analysis of Section 3.3 separately for the pre- and post publication period. Specifically, we construct portfolios at each point in time only using factors before and after their publication date, respectively. Panel A of Table 8 shows the pre publication multifactor sorting results. The high minus low (H-L) return of naively timed portfolio sorts, i.e. sorting on the unconditional mean, is even higher when conditioning on pre publication periods than over the full sample (Table 5). The PLS1-timed version of the multifactor portfolio again yields a higher average return in the pre period. In Panel B of Table 8 we show the multifactor portfolio sorts for the post publication sample period. The naively timed factor portfolios yield considerably lower returns in the post publication period than in the pre period (7.8% vs. 12.7% for the "H-L" portfolio). The "H-L" portfolio of the PLS1-timed factors shows almost no drop in return, albeit a small increase in volatility. This analysis shows that the gains of factor timing have increased in the post publication period, whereas the gains of passive factor investing have decreased.

Table 8: Portfolio Sorts Based on Predicted Factor Returns

This table shows pre- and post-publication performance statistics for sorted portfolios based on the predicted factor returns. At the end of each month t we sort factors into 5 portfolios based on their $t+1$ predicted return and construct a high-low (H-L) portfolio. For naively timed factors, portfolios are sorted based on the historical average return. PLS1 timed factor sorts are based on predictions from partial least squares regressions with 1 component. In Panel A (Panel B), we condition of factors for which month t is in the pre-publication (post publication) period. In subpanels i, we report the annualized mean predicted return (Pred), realized return (R), standard deviation (SD), Sharpe ratio (SR) and maximum drawdown (maxDD). Subpanels ii shows regression results, where we regress returns on the CAPM and FF5+Momentum asset pricing models. We report average t-statistics in brackets. Subpanels iii displays the timing success. α, ΔR, and ΔSR exhibit the time-series regression alpha, difference in return and difference in Sharpe ratio of the timed vs. untimed factor returns. We again report t- and z-statistics in brackets. We describe the factors, their publication date and their allocation to an economic category in Table A.1. Table B.1 describes the timing signals.

	Naively timed factors						PLS1 timed factors					
	L	2	3	4	H	H-L	L	2	3	4	H	H-L
A. Pre-publication												
i. Portfolio performance												
Pred							-4.833	1.599	4.988	8.561	16.405	21.238
R	1.321	4.900	5.25	6.729	12.106	10.785	-3.471	3.486	6.914	8.463	15.249	18.720
SD	6.382	3.907	4.155	5.786	8.082	12.726	7.130	3.844	4.335	6.012	8.867	14.824
SR	0.207	1.254	1.263	1.163	1.498	0.847	-0.487	0.907	1.595	1.408	1.720	1.263
maxDD	18.497	10.044	8.668	13.653	16.250	32.461	80.863	11.669	10.803	14.063	18.209	36.345
ii. Regression results												
CAPM α	0.770	5.043	5.859	7.728	12.819	12.049	-3.885	3.860	7.565	9.027	16.014	19.899
	0.799	8.394	9.754	9.505	10.514	6.316	-3.565	6.688	12.116	9.984	11.959	8.859
FF5 + MOM α	1.774	5.068	4.030	4.779	8.157	6.383	-1.935	3.175	5.714	5.453	11.720	13.655
	1.953	8.193	7.368	6.456	7.902	3.899	-1.856	5.340	9.615	6.796	9.869	6.777
iii. Timing success												
α							-4.355	1.711	3.318	3.481	5.297	10.108
							(-4.968)	(2.938)	(6.221)	(5.105)	(5.402)	(5.936)
ΔR							-4.792	-1.414	1.664	1.734	3.143	7.935
							(-5.142)	(-2.119)	(3.086)	(2.530)	(3.395)	(4.672)
ΔSR							-0.694	-0.347	0.332	0.245	0.222	0.415
							(-5.014)	(-1.952)	(2.441)	(1.992)	(1.873)	(3.281)
B. Post-publication												
i. Portfolio performance												
Pred							-6.229	0.799	4.380	8.431	17.623	23.851
R	0.405	0.375	1.897	4.464	8.254	7.849	-5.850	1.229	1.988	6.459	12.549	18.399
SD	8.809	7.271	7.403	6.947	7.972	13.545	9.832	6.948	5.461	8.271	9.812	16.791
SR	0.046	0.052	0.256	0.643	1.035	0.579	-0.595	0.177	0.364	0.781	1.279	1.096
maxDD	55.543	53.23	46.772	26.492	20.756	39.131	93.422	36.568	45.722	23.405	12.503	37.000
ii. Regression results												
CAPM α	-0.181	0.646	2.121	5.710	9.837	10.018	-6.567	1.710	2.544	7.678	13.799	20.365
	-0.132	0.566	1.824	5.530	8.416	4.916	-4.284	1.577	3.013	6.127	9.191	7.892
FF5 + MOM α	-0.549	0.714	0.335	3.919	6.622	7.171	-6.129	0.951	1.324	4.616	11.507	17.636
	-0.424	0.616	0.284	3.869	6.270	3.805	-4.287	0.860	1.543	3.832	7.803	7.242
iii. Timing success												
α							-6.126	1.108	1.497	4.468	7.280	13.311
							(-5.059)	(1.090)	(1.878)	(3.687)	(5.351)	(5.903)
ΔR							-6.255	0.853	0.090	1.995	4.295	10.550
							(-4.865)	(0.671)	(0.078)	(1.498)	(3.119)	(4.506)
ΔSR							-0.641	0.125	0.108	0.138	0.244	0.516
							(-4.629)	(0.700)	(0.607)	(0.785)	(1.535)	(3.320)

3.5 Stock-level Timing Portfolios

In all of the previous analyses we have taken factor portfolios as primitives. Since the factors are zero net investment portfolios, combinations of them will of course also be zero net investment portfolios and the results can be interpreted as proper excess returns. Nonetheless, it is beneficial to take a look “under the hood” to get more insights into the properties of multifactor timing portfolios for multiple reasons. To properly assess turnover of factor timing strategies, we need to compute the actual positions for each security in the portfolio, as the same stock may be in the long leg of one factor portfolio and in the short leg of another portfolio. When implementing dynamic investment strategies in real-world portfolios, investors will clearly transact only on the difference between the desired net position and the current actual holdings. DeMiguel, Martin-Utrera, Nogales, and Uppal (2020) show that many trades cancel out when multiple factors are combined into one portfolio. A second important reason is due to real life frictions and constraints investors are facing. For example, leverage or short-sale constraints may inhibit the implementation of the optimal timing portfolio. The only way to gain more insight into these issues is to unpack the timing portfolio and study the multifactor timing portfolios at the individual security level.

To keep track of the net position of stock i in a multifactor timing portfolio, we derive the aggregate weight $w_{i,t}$ by aggregating across the $j = 1, ..., N$ factors:

$$w_{i,t} = \sum_{j=1}^{N} w_{i,j,t} \,, \tag{13}$$

where $w_{i,j,t}$ is firm i's weight in factor j at time t. We then avoid short positions in individual stocks, and only consider those stocks which receive a positive aggregate weight,

$$w_{i,t}^{+} = \max\left[0, w_{i,t}\right] \tag{14}$$

and normalize the sum of weights to one. Similarly, we derive stock-level weights $w_{i,t}^{PLS,+}$ from the timed factor portfolios.

We define the monthly turnover as the change in weights $TO_t = \frac{1}{2}\sum_i \left|w_{i,t} - w_{i,t}^{bh}\right|$, where $w_{i,t}$ is the weight of firm i at time t, $w_{i,t}^{bh}$ is the buy and hold weight, i.e. the weight of firm i at time t when no action is taken on the previous period's weight $w_{i,t-1}$. We define $w_{i,t}^{bh}$ as $w_{i,t}^{bh} = \frac{mcap_{t-1} w_{i,t-1}\left(1+r_{i,t}^{exd}\right)}{mcap_t}$, where $mcap_t$ is the market capitalization of the entire investment universe at time t. Note that this can change from $t-1$ to t not only because of performance, but also because of IPOs, SEOs, buybacks, and dividend payments. $r_{i,t}^{exd}$ is the return of firm i excluding dividends from $t-1$ to t.

Table 9 shows the results. Panels A and B report results for small and large-capitalization stocks, respectively, where we split the sample using the median NYSE market equity at the end of June of year t (see Fama and French, 1992).

Table 9: Individual Stock-level Timing Portfolios

For this table, timed portfolios are constructed from individual securities rather than from factors. To this end we aggregate the underlying security weights (long and short) from all timed factor portfolios. We then obtain portfolios that consist of those stocks that have positive aggregate weights. Panels A and B report results for small and large-capitalization stocks in the CRSP universe, respectively, where we split the sample in June of year t using the median NYSE market equity and keep firms from July of year t to June of year $t+1$. ALL_VW is the value-weighted portfolio return of small and large cap stocks, respectively. Untimed refers to portfolio weights based on the original factor definition. PLS1 timed shows portfolio timing based on partial least squares regressions with a single component. We report annualized mean return (R), standard deviation (SD), Sharpe ratio (SR), maximum drawdown (maxDD), average number of firms in the portfolio (N), and annualized turnover (Turn). The sample period is January 1974 to December 2020. We describe the factors and their allocation into an economic category in Table A.1.

	R	SD	SR	maxDD	N	Turn
A. Small capitalization stocks						
ALL_VW	12.832	20.420	0.413	55.076	3,945	7.053
Untimed	24.119	21.684	0.910	57.828	2,216	286.435
PLS1 timed	25.611	22.663	0.936	50.035	2,130	401.917
B. Large capitalization stocks						
ALL_VW	9.265	15.538	0.314	51.585	929	3.484
Untimed	12.092	16.142	0.477	49.111	341	312.770
PLS1 timed	13.506	17.359	0.525	51.084	377	455.107

We find several interesting results. First, there is a tremendous gain in portfolio performance relative to the market weighted return, even when we just use untimed factors to form portfolios. When we restrict the sample to small stocks, the annualized return of the untimed portfolio is about 11% p.a. higher than the benchmark, which constitutes an increase of roughly 80%. Results for large-capitalization firms suggest a smaller, but still high absolute (3%) and relative (25%) over-performance. This increase in performance is unmatched by the increase in portfolio risk. Even though the standard deviation increases in all groups, the rise is less pronounced than the return, resulting in much larger Sharpe ratios. The Sharpe ratio for the small (large) sample rises from 0.413 (0.314) to 0.910 (0.477), which is an increase of 120% (50%).

Second, our timing model, denoted as PLS1, further increases the performance and risk-adjusted returns. The small cap portfolio yields an annualized return of 25.6% with a Sharpe ratio of 0.94. Alongside the impressive gain in performance, we find decreasing maximum drawdowns and a reasonable number of firms in the portfolio. However, timing and re-balancing on a monthly basis results in high turnover of roughly 400% per year.

There is merit in focusing on the best in class firms, i.e. firms that have the largest weights across all timed factors. We provide insights in the appendix, Table C.3. We use a subset of firms in each size sample, retaining only firms with weights above the median and in the top quintile, respectively. Generally speaking, these portfolios have higher returns and higher Sharpe ratios, but also slightly higher standard deviations. The increase in standard deviation might be due to an increase in idiosyncratic risk, because of the lower number of holdings in the respective portfolios. For example, in the large-cap sample, the number of firms is on average 189, when we just use firms in the upper half of the weight distribution, and about 76 when we use the highest quintile. Interestingly, we find that these portfolios generate much lower turnover than the base-case PLS1 portfolio. This suggests that firms have, on aggregate across all factors, relatively sticky weights. The strategy that focuses on large-cap stocks with a weight in the top 50% resulting from timing with PLS1, increases the Sharpe ratio by roughly 80% to 0.53 (relative to 0.31 for the market-weight CRSP large-cap universe). With an average number of 190 large-cap stocks in the portfolio and a turnover of 340%, the resulting strategy can very likely be implemented in practice.

3.6 Performance in Different Economic Regimes

Next, we analyze the persistence of results across different economic regimes. We split the data along two dimensions. First we split the sample by the implied market volatility, i.e. the CBOE S&P 500 volatility index, into high VIX regimes when the VIX at month t is above the historical median, and vice versa. The number of observations is 164 and 207, respectively. Second, we provide statistics for NBER recessions and expansions, with 73 and 492 observations, respectively. Table 10 shows the results. Panel A shows that factor timing works equally well in both high and low volatility regimes. Average returns of PLS1-timed portfolios exceed the returns from untimed factor-based portfolios, which in turn are higher than market-capitalization weighted portfolios. The difference between the two former strategies amounts to approximately 2% p.a. for both small and large capitalization stocks, and irrespective of the VIX regime. There is one exception to the persistent outperformance when comparing Sharpe ratios: The Sharpe ratio of the PLS1-timed small-cap portfolio is slightly below the untimed factor portfolio (1.01 vs. 1.04). Panel B reveals performance statistics during economic turmoil. Most notably, when the economy is in a recession, the return for the sample of small (large) stocks is -11.7% (-12.4%). However, the PLS1 timing model does improve the return tremendously. Small (large) capitalization stock portfolios return roughly 13 (6) percent above and beyond the market, and 3 (1) percent above the untimed factor portfolios. This result is not dwarfed when we investigate the performance during expansions. Here the PLS1 timing portfolio again provides the highest outperformance, with returns being 12 (4) percent above the small and large market portfolios, respectively, and 1 (1) percent above the small and large cap untimed factor-based portfolios.

Table 10: Performance of Stock-level Timing Portfolios During Crises

This table shows performance statistics for high (above the historical median) and low (below historical median) implied volatility (i.e. CBOE S&P 500 volatility index, VIX) regimes, and NBER recession regimes for long-only equity portfolios. We aggregate all underlying security weights from all timed factor portfolios. We then retain only stocks that have positive aggregate weights. Panels A and B report results conditional on the VIX regime, for small and large-capitalization stocks, respectively. We split the sample in June of year t using the median NYSE market equity and keep firms from July of year t to June of year $t+1$. Panels B and C report results conditional on recession regimes. ALL_VW is the value-weighted return for small and large cap stocks, respectively. Untimed refers to portfolio weights based on the original factor definition. PLS1 shows portfolio timing based on partial least squares regressions with a single component. We report annualized mean return (R), standard deviation (SD), and Sharpe ratio (SR). The sample period is January 1990 to December 2020 for VIX regimes and January 1974 to December 2020 for recession regimes. We describe the factors and their allocation to an economic category in Table A.1.

	High VIX (N=164)			*Low VIX (N=207)*		
	R	SD	SR	R	SD	SR
A. Small capitalization stocks						
ALL_VW	14.192	17.940	0.426	13.783	22.002	0.369
Untimed	25.879	19.230	1.005	29.394	22.845	1.039
PLS1 timed	27.869	19.947	1.069	31.329	25.521	1.006
B. Large capitalization stocks						
ALL_VW	10.589	15.672	0.258	8.251	16.058	0.161
Untimed	12.905	15.691	0.405	12.550	16.001	0.431
PLS1 timed	14.391	16.708	0.469	14.543	18.424	0.482
	NBER recession (N=73)			*NBER expansion (N=492)*		
	R	SD	SR	R	SD	SR
C. Small capitalization stocks						
ALL_VW	-11.742	29.598	-0.608	16.479	18.492	0.669
Untimed	0.143	33.266	-0.184	27.677	19.226	1.226
PLS1 timed	3.481	30.685	-0.091	28.894	21.094	1.175
D. Large capitalization stocks						
ALL_VW	-12.382	22.780	-0.818	12.477	13.949	0.600
Untimed	-9.835	24.123	-0.667	15.346	14.392	0.780
PLS1 timed	-9.197	24.306	-0.636	16.874	15.880	0.804

4 Additional Empirical Results

4.1 Factor or Market Timing

In the previous sections, we provide strong evidence for factor timing using signals constructed separately for each individual factor. One might wonder if the outperformance of timed factors truly comes from *factor timing* or if it could alternatively be explained by *market timing*. To answer this question, we run regressions in the style of Henriksson and Merton (1981). Specifically, we regress for each factor its timed returns onto the market excess return and the maximum of zero and the market excess return, i.e.

$$f^{\tau}_{i,t} = \alpha + \beta r_{m,t} + \gamma r^{+}_{m,t} + \varepsilon_t \, , \tag{15}$$

where $f^{\tau}_{i,t}$ denotes the timed factor return using PLS1, $r_{m,t}$ is the excess return on the market and $r^{+}_{m,t} := \max\{r_{m,t}, 0\}$. While a strategy that earns the maximum of the market return and zero cannot be implemented in real time, it sheds light into whether observed outperformance is due to alpha or time-varying market exposure (i.e., market timing). Table 11 displays the results. Panel A reports the mean and quartiles of the coefficients α, β, and γ for the full cross-section of factors. The average t-statistic of alphas from this regression exceeds the average t-statistic of the alphas obtained from regressing timed factor returns on untimed factor returns, as shown in Table 3, Panel C. In contrast, the coefficients γ on the maximum of the market return and zero are basically zero across the entire cross-section. Panel B reports the average coefficients and their t-statistics when the factors are split into economic categories. Hence, there is no indication that market timing plays a role in explaining timed factor returns.

Table 11: Henriksson and Merton (1981) Market Timing Regression

This table shows statistics for the coefficient estimates of a timing regression. For each factor, we regress its returns onto the market excess return and the maximum of zero and the market excess return, i.e. $f_{i,t}^{j} = \alpha + \beta r_{m,t} + \gamma r_{m,t}^{+} + \varepsilon_t$. $f_{i,t}^{j}$ denotes the timed factor return, $r_{m,t}$ is the excess return on the market and $r_{m,t}^{+} := \max\{r_{m,t}, 0\}$. The timing regression is carried out for each factor. Alphas are annualized. Panel A shows the mean, the 25%, 50% and 75% quantile of the coefficient estimates. In Panel B, we show the average coefficient and t-statistics in parentheses for each economic category. We describe the factors and their allocation to an economic category in Table A.1.

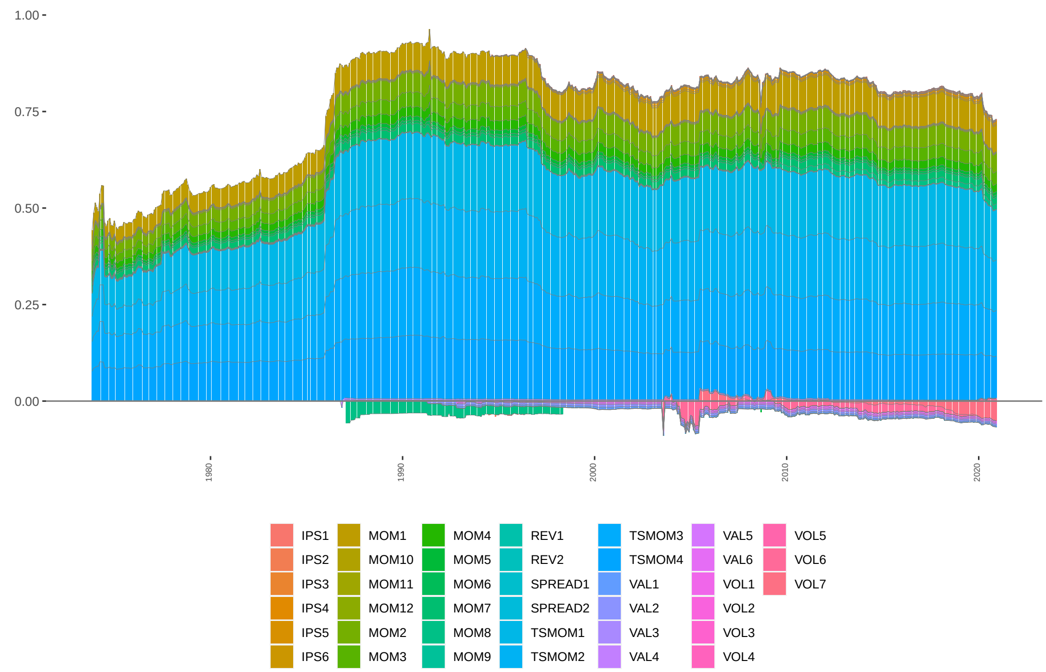
	(1) α	(2) β	(3) γ
A. Full sample			
Mean	0.059	-0.099	0.031
	(1.814)	(-1.329)	(0.150)
25 perc.	0.019	-0.182	-0.063
	(0.700)	(-2.528)	(-0.664)
50 perc.	0.047	-0.070	0.022
	(1.595)	(-1.141)	(0.216)
75 perc.	0.087	0.008	0.131
	(2.674)	(0.147)	(1.037)
B. Economic category			
Intangibles	0.027	-0.058	0.069
	(1.222)	(-1.162)	(0.277)
Investment	0.054	-0.047	0.005
	(2.285)	(-0.934)	(0.002)
Momentum	0.120	-0.087	-0.121
	(3.149)	(-0.808)	(-0.859)
Profitability	0.069	-0.170	0.076
	(1.561)	(-1.743)	(0.491)
Trading frictions	0.056	-0.226	0.057
	(1.737)	(-2.916)	(0.272)
Value vs. growth	0.069	-0.114	0.039
	(2.060)	(-1.479)	(0.216)
Other	0.060	-0.044	0.025
	(1.566)	(-0.596)	(0.183)

4.2 Importance of Timing Signals

Using partial least squares, we combine 39 signals into one aggregated timing signal. It is interesting to understand to which degree individual signals contribute to the combined signal, and if the importance of an individual signal changes over time. To illustrate the relative importance of timing signals, at each point in time and for each factor portfolio, we obtain the loadings of each signal in the first PLS component. We then take a cross-sectional average across all factor portfolios. As the PLS loadings are only identified up to rotation, we normalize through dividing by the time-series maximum to achieve a maximum of one over the full sample. The resulting normalized loadings indicate the relative importance of different timing signals over time. Figure 5 shows that momentum and time series momentum signals dominate the aggregate signal. Volatility signals contribute to a lesser degree, and with a negative sign.

Figure 5: Relative Importance of Timing Signals (PLS Loadings)

This figure shows the relative importance of the timing signal for the partial least squares (PLS) factor. At each time we take a cross-sectional average. As the PLS loadings are only identified up to rotation, we divide by the time-series maximum to achieve a maximum of one over the full sample. We describe the factors and their allocation to an economic category in Table A.1. Table B.1 describes the timing signals.



4.3 Timing persistence

Timing effects may arise due to liquidity effects, i.e. rather than capturing time variation in prices of risk, the timing effects might simply capture the returns to liquidity provision. We therefore study the persistence of factor timing. Table 12 shows the returns to univariate factor timing for holding periods of up to one year. The results show that for the average factor, the timed return is greater than its untimed counterpart for a holding period of up to one year. The results are however stronger at shorter horizons.

Table 12: Univariate Factor Timing with Longer Holding Horizon

This table shows results for univariate factor timing portfolios with longer holding periods. For each factor i, the predicted returns ($\hat{f}_{i,t+1}$) are from partial least squares (PLS) regressions, where the number of components equals 1. The timed portfolio, denoted by $f^{\tau}_{i,t+1->t+T}$, is equal to the original factor return, when the forecast is positive. In the event of a negative predicted return, the timed factor return is essentially its inverse, i.e. the short-long untimed portfolio return. T, with $T \in \{1, 3, 6, 12\}$, denotes the holding period horizon (in months). All statistics are obtained in a two-step procedure: First, we compute statistics for each individual factor separately for its out-of-sample period. Second, we take cross-sectional averages. Panel A reports the annualized average return in percent, annualized standard deviation in percent, annualized Sharpe ratio, and maximum drawdown in percent. Panel B shows regression results, where we regress returns on the CAPM and FF5+Momentum asset pricing models. We report average t-statistics in parenthesis. Panel C displays the timing success. α, ΔR, and ΔSR denote the time-series regression alpha, difference in return and difference in Sharpe ratio of the timed vs. untimed factor returns. We again report t- and z-statistics in parenthesis. % $\hat{f}_{i,t+1} > 0$ (% $\hat{f}_{i,t+1} < 0$) shows the fraction of positive (negative) predictions. $\hat{f}_{i,t+1} > 0 \mid f_{i,t+1} > 0$ ($\hat{f}_{i,t+1} < 0 \mid f_{i,t+1} < 0$) reports the fraction of positive (negative) return prediction conditional on the untimed factor returns' $f_{i,t+1}$ realisation being positive (negative). hit rate reports the fraction where the sign of the predicted return $\hat{f}_{i,t+1}$ corresponds to the sign of $f_{i,t+1}$. We describe the factors, their publication date and their allocation to an economic category in Table A.1. Table B.1 describes the timing signals.

	Untimed factors	PLS1 - timed factors			
N of months holding period		1	3	6	12
A. Performance					
R	4.040	5.879	5.091	4.869	4.421
SD	12.782	12.752	12.776	12.783	12.787
SR	0.336	0.482	0.426	0.409	0.385
maxDD	45.963	37.881	40.73	41.922	43.366
B. Regression results					
CAPM α	4.515	6.588	5.852	5.585	5.109
	(2.179)	(3.042)	(2.762)	(2.659)	(2.539)
FF5 + MOM α	3.368	4.938	4.054	3.857	3.795
	(1.950)	(2.497)	(2.171)	(2.114)	(2.170)
C. Timing success					
α		2.354	1.375	0.994	0.313
		(1.112)	(0.667)	(0.444)	(0.172)
ΔR		1.840	1.051	0.829	0.381
		(0.490)	(0.232)	(0.111)	(0.017)
ΔSR		0.147	0.091	0.073	0.049
		(0.481)	(0.223)	(0.103)	(0.012)
% $\hat{f}_{i,t+1} > 0$		74.366	76.75	78.173	79.772
% $\hat{f}_{i,t+1} < 0$		25.634	23.25	21.827	20.228
$\hat{f}_{i,t+1} > 0 \mid f_{i,t+1} > 0$		79.025	80.611	81.686	83.004
$\hat{f}_{i,t+1} < 0 \mid f_{i,t+1} < 0$		29.905	26.287	24.329	22.319
hit rate		56.930	56.174	55.886	55.706

In Figure C.1 in the appendix, we break up the results by economic categories. The figure shows that up to a two months horizon is possible in all categories except momentum. In summary, the analysis reveals that in most categories timing is possible even at longer horizons.

5 Conclusion

The academic literature has identified many asset pricing factors – the *factor zoo*. It has also analyzed whether risk premia associated with these factors are time-varying and whether it is possible to successfully time investors' exposure to the various risk factors. The evidence on the latter question is inconclusive, as different papers have focused on very different sets of factors and predictive variables. In this paper we conduct a comprehensive analysis of factor timing, simultaneously considering a large set of risk factors and of prediction variables. Our analysis reveals that factor timing is indeed possible. Predictability is not concentrated in short subsamples of the data and does not decay in recent time periods. In short, factor risk premia are robustly predictable. Our evidence reveals that factor timing is beneficial to investors relative to passive "harvesting" of risk premia. Whereas the average premia for factors have decreased post publication, the gains of factor timing have not. To the contrary, the gains from factor timing have increased in the post publication period.

In addition, our results have important implications for asset pricing theories and models. Our results show that there is a large difference between the conditional and unconditional behavior of factor returns and risk premia. In particular, models of constant conditional risk premia appear inconsistent with the data. Our findings are also useful for the design of models of the stochastic discount factor, in particular, models that imply i.i.d. innovations of the SDF cannot match our empirical findings and are likely to be rejected in the data.

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Appendices

A Anomalies

This section describes the details of our dataset. As outlined in Section 2, our dataset has been derived from the open source code provided by Chen and Zimmermann (2022). It encompasses over 300 equity portfolios that have been constructed by sorting firms on various characteristics. The sample includes NYSE, AMEX, and Nasdaq ordinary common stocks for the time period from 1926 to 2020. The list of firm characteristics can be constructed from CRSP, Compustat, and IBES, FRED data. Multiple characteristics require specific data to reconstruct the results of the original studies, which are readily available on the authors' websites. For each characteristic, Chen and Zimmermann (2022) replicate portfolios used in the original papers that introduced the anomaly in the literature. Table A.1 displays a brief description of the firm characteristics.

Table A.1: Summary of Anomaly Variables

Acronym	Description	Author	Year/Month	Journal	Economic category
AbnormalAccruals	Abnormal Accruals	Xie	2001/7	AR	Investment
AbnormalAccrualsPercent	Percent Abnormal Accruals	Hafzalla, Lundholm, Van Winkle	2011/1	AR	Investment
AccrualQuality	Accrual Quality	Francis, LaFond, Olsson, Schipper	2005/6	JAE	Investment
AccrualQualityJune	Accrual Quality in June	Francis, LaFond, Olsson, Schipper	2005/6	JAE	Investment
Accruals	Accruals	Sloan	1996/7	AR	Investment
AccrualsBM	Book-to-market and accruals	Bartov and Kim	2004/12	RFQA	Investment
Activism1	Takeover vulnerability	Cremers and Nair	2005/12	JF	Other
Activism2	Active shareholders	Cremers and Nair	2005/12	JF	Intangibles
AdExp	Advertising Expense	Chan, Lakonishok and Sougiannis	2001/12	JF	Intangibles
AgeIPO	IPO and age	Ritter	1991/3	JF	Intangibles
AM	Total assets to market	Fama and French	1992/6	JF	Value vs. growth
AMq	Total assets to market (quarterly)	Fama and French	1992/6	JF	Value vs. growth
AnalystRevision	EPS forecast revision	Hawkins, Chamberlin, Daniel	1984/9	FAJ	Momentum
AnalystValue	Analyst Value	Frankel and Lee	1998/6	JAE	Intangibles
AnnouncementReturn	Earnings announcement return	Chan, Jegadeesh and Lakonishok	1996/12	JF	Momentum
AOP	Analyst Optimism	Frankel and Lee	1998/6	JAE	Intangibles
AssetGrowth	Asset growth	Cooper, Gulen and Schill	2008/8	JF	Investment
AssetGrowth-q	Asset growth quarterly	Cooper, Gulen and Schill	2008/8	JF	Investment
AssetLiquidityBook	Asset liquidity over book assets	Ortiz-Molina and Phillips	2014/2	JFQA	Other
AssetLiquidityBookQuart	Asset liquidity over book (qtrly)	Ortiz-Molina and Phillips	2014/2	JFQA	Other
AssetLiquidityMarket	Asset liquidity over market	Ortiz-Molina and Phillips	2014/2	JFQA	Other
AssetLiquidityMarketQuart	Asset liquidity over market (qtrly)	Ortiz-Molina and Phillips	2014/2	JFQA	Other
AssetTurnover	Asset Turnover	Soliman	2008/5	AR	Other
AssetTurnover-q	Asset Turnover	Soliman	2008/5	AR	Other
Beta	CAPM beta	Fama and MacBeth	1973/5	JPE	Trading frictions
BetaBDLeverage	Broker-Dealer Leverage Beta	Adrian, Etula and Muir	2014/12	JF	Trading frictions
betaCC	Illiquidity-illiquidity beta (beta2i)	Acharya and Pedersen	2005/4	JFE	Trading frictions
betaCR	Illiquidity-market return beta (beta4i)	Acharya and Pedersen	2005/4	JFE	Trading frictions
BetaDimson	Dimson Beta	Dimson	1979/6	JFE	Trading frictions
BetaFP	Frazzini-Pedersen Beta	Frazzini and Pedersen	2014/1	JFE	Trading frictions
BetaLiquidityPS	Pastor-Stambaugh liquidity beta	Pastor and Stambaugh	2003/6	JPE	Trading frictions
betaNet	Net liquidity beta (betanet,p)	Acharya and Pedersen	2005/4	JFE	Trading frictions
betaRC	Return-market illiquidity beta	Acharya and Pedersen	2005/4	JFE	Trading frictions
betaRR	Return-market return illiquidity beta	Acharya and Pedersen	2005/4	JFE	Trading frictions
BetaSquared	CAPM beta squred	Fama and MacBeth	1973/5	JPE	Trading frictions
BetaTailRisk	Tail risk beta	Kelly and Jiang	2014/10	RFS	Trading frictions
betaVIX	Systematic volatility	Ang et al.	2006/2	JF	Trading frictions
BidAskSpread	Bid-ask spread	Amihud and Mendelsohn	1986/12	JFE	Trading frictions
BM	Book to market using most recent ME	Rosenberg, Reid, and Lanstein	1985/4	JF	Value vs. growth
BMdec	Book to market using December ME	Fama and French	1992/6	JPM	Value vs. growth
BMq	Book to market (quarterly)	Rosenberg, Reid, and Lanstein	1985/4	JF	Value vs. growth
BookLeverage	Book leverage (annual)	Fama and French	1992/6	JF	Value vs. growth
BookLeverageQuarterly	Book leverage (quarterly)	Fama and French	1992/6	JF	Value vs. growth
BPEBM	Leverage component of BM	Penman, Richardson and Tuna	2007/5	JAR	Value vs. growth
BrandCapital	Brand capital to assets	Belo, Lin and Vitorino	2014/1	RED	Intangibles
BrandInvest	Brand capital investment	Belo, Lin and Vitorino	2014/1	RED	Intangibles
CapTurnover	Capital turnover	Haugen and Baker	1996/7	JFE	Other
CapTurnover-q	Capital turnover (quarterly)	Haugen and Baker	1996/7	JFE	Other
Cash	Cash to assets	Palazzo	2012/4	JFE	Value vs. growth
cashdebt	CF to debt	Ou and Penman	1989/NA	JAR	Other
CashProd	Cash Productivity	Chandrashekar and Rao	2009/1	WP	Intangibles
CBOperProf	Cash-based operating profitability	Ball et al.	2016/7	JFE	Profitability
CBOperProfLagAT	Cash-based oper prof lagged assets	Ball et al.	2016/7	JFE	Profitability
CBOperProfLagAT-q	Cash-based oper prof lagged assets qtrly	Ball et al.	2016/7	JFE	Profitability
CF	Cash flow to market	Lakonishok, Shleifer, Vishny	1994/12	JF	Value vs. growth
cfp	Operating Cash flows to price	Desai, Rajgopal, Venkatachalam	2004/4	AR	Value vs. growth
cfpq	Operating Cash flows to price quarterly	Desai, Rajgopal, Venkatachalam	2004/4	AR	Value vs. growth
CFq	Cash flow to market quarterly	Lakonishok, Shleifer, Vishny	1994/12	JF	Value vs. growth
ChangeInRecommendation	Change in recommendation	Jegadeesh et al.	2004/6	JF	Intangibles
ChangeRoA	Change in Return on assets	Balakrishnan, Bartov and Faurel	2010/5	NA	Profitability
ChangeRoE	Change in Return on equity	Balakrishnan, Bartov and Faurel	2010/5	NA	Profitability
ChAssetTurnover	Change in Asset Turnover	Soliman	2008/5	AR	Profitability
ChEQ	Growth in book equity	Lockwood and Prombutr	2010/12	JFR	Intangibles
ChForecastAccrual	Change in Forecast and Accrual	Barth and Hutton	2004/3	RAS	Intangibles
ChInv	Inventory Growth	Thomas and Zhang	2002/6	RAS	Investment
ChInvIA	Change in capital inv (ind adj)	Abarbanell and Bushee	1998/1	AR	Investment
ChNAnalyst	Decline in Analyst Coverage	Scherbina	2008/NA	ROF	Intangibles
ChNCOA	Change in Noncurrent Operating Assets	Soliman	2008/5	AR	Investment
ChNCOL	Change in Noncurrent Operating Liab	Soliman	2008/5	AR	Investment
ChNNCOA	Change in Net Noncurrent Op Assets	Soliman	2008/5	AR	Investment

Table A.1 – cont.

Acronym	Description	Author	Year/Month	Journal	Economic category
ChNWC	Change in Net Working Capital	Soliman	2008/5	AR	Profitability
ChPM	Change in Profit Margin	Soliman	2008/5	AR	Other
ChTax	Change in Taxes	Thomas and Zhang	2011/6	JAR	Intangibles
CitationsRD	Citations to RD expenses	Hirschleifer, Hsu and Li	2013/3	JFE	Other
CompEquIss	Composite equity issuance	Daniel and Titman	2006/8	JF	Investment
CompositeDebtIssuance	Composite debt issuance	Lyandres, Sun and Zhang	2008/11	RFS	Investment
ConsRecomm	Consensus Recommendation	Barber et al.	2002/4	JF	Other
ConvDebt	Convertible debt indicator	Valta	2016/2	JFQA	Intangibles
CoskewACX	Coskewness using daily returns	Ang, Chen and Xing	2006/12	RFS	Trading frictions
Coskewness	Coskewness	Harvey and Siddique	2000/6	JF	Trading frictions
CredRatDG	Credit Rating Downgrade	Dichev and Piotroski	2001/2	JF	Profitability
currat	Current Ratio	Ou and Penman	1989/NA	JAR	Value vs. growth
CustomerMomentum	Customer momentum	Cohen and Frazzini	2008/8	JF	Other
DebtIssuance	Debt Issuance	Spiess and Affleck-Graves	1999/10	JFE	Investment
DelayAcct	Accounting component of price delay	Callen, Khan and Lu	2013/3	CAR	Other
DelayNonAcct	Non-accounting component of price delay	Callen, Khan and Lu	2013/3	CAR	Other
DelBreadth	Breadth of ownership	Chen, Hong and Stein	2002/11	JFE	Intangibles
DelCOA	Change in current operating assets	Richardson et al.	2005/9	JAE	Investment
DelCOL	Change in current operating liabilities	Richardson et al.	2005/9	JAE	Investment
DelDRC	Deferred Revenue	Prakash and Sinha	2012/2	CAR	Profitability
DelEqu	Change in equity to assets	Richardson et al.	2005/9	JAE	Investment
DelFINL	Change in financial liabilities	Richardson et al.	2005/9	JAE	Investment
DelLTI	Change in long-term investment	Richardson et al.	2005/9	JAE	Investment
DelNetFin	Change in net financial assets	Richardson et al.	2005/9	JAE	Investment
DelSTI	Change in short-term investment	Richardson et al.	2005/9	JAE	Investment
depr	Depreciation to PPE	Holthausen and Larcker	1992/6	JAE	Other
DivInit	Dividend Initiation	Michaely, Thaler and Womack	1995/6	JF	Value vs. growth
DivOmit	Dividend Omission	Michaely, Thaler and Womack	1995/6	JF	Value vs. growth
DivSeason	Dividend seasonality	Hartzmark and Salomon	2013/9	JFE	Value vs. growth
DivYield	Dividend yield for small stocks	Naranjo, Nimalendran, Ryngaert	1998/12	JF	Value vs. growth
DivYieldAnn	Last year's dividends over price	Naranjo, Nimalendran, Ryngaert	1998/12	NA	Value vs. growth
DivYieldST	Predicted div yield next month	Litzenberger and Ramaswamy	1979/6	JF	Value vs. growth
dNoa	change in net operating assets	Hirshleifer, Hou, Teoh, Zhang	2004/12	JAE	Investment
DolVol	Past trading volume	Brennan, Chordia, Subra	1998/9	JFE	Trading frictions
DownRecomm	Down forecast EPS	Barber et al.	2002/4	JF	Intangibles
DownsideBeta	Downside beta	Ang, Chen and Xing	2006/12	RFS	Trading frictions
EarningsConservatism	Earnings conservatism	Francis, LaFond, Olsson, Schipper	2004/10	AR	Other
EarningsConsistency	Earnings consistency	Alwathainani	2009/9	BAR	Intangibles
EarningsForecastDisparity	Long-vs-short EPS forecasts	Da and Warachka	2011/5	JFE	Intangibles
EarningsPersistence	Earnings persistence	Francis, LaFond, Olsson, Schipper	2004/10	AR	Other
EarningsPredictability	Earnings Predictability	Francis, LaFond, Olsson, Schipper	2004/10	AR	Other
EarningsSmoothness	Earnings Smoothness	Francis, LaFond, Olsson, Schipper	2004/10	AR	Other
EarningsStreak	Earnings surprise streak	Loh and Warachka	2012/7	MS	Other
EarningsSurprise	Earnings Surprise	Foster, Olsen and Shevlin	1984/10	AR	Momentum
EarningsTimeliness	Earnings timeliness	Francis, LaFond, Olsson, Schipper	2004/10	AR	Other
EarningsValueRelevance	Value relevance of earnings	Francis, LaFond, Olsson, Schipper	2004/10	AR	Other
EarnSupBig	Earnings surprise of big firms	Hou	2007/7	RFS	Momentum
EBM	Enterprise component of BM	Penman, Richardson and Tuna	2007/5	JAR	Value vs. growth
EBM-q	Enterprise component of BM	Penman, Richardson and Tuna	2007/5	JAR	Value vs. growth
EntMult	Enterprise Multiple	Loughran and Wellman	2011/12	JFQA	Value vs. growth
EntMult-q	Enterprise Multiple quarterly	Loughran and Wellman	2011/12	JFQA	Value vs. growth
EP	Earnings-to-Price Ratio	Basu	1977/6	JF	Value vs. growth
EPq	Earnings-to-Price Ratio	Basu	1977/6	JF	Value vs. growth
EquityDuration	Equity Duration	Dechow, Sloan and Soliman	2004/6	RAS	Value vs. growth
ETR	Effective Tax Rate	Abarbanell and Bushee	1998/1	AR	Other
ExchSwitch	Exchange Switch	Dharan and Ikenberry	1995/12	JF	Trading frictions
ExclExp	Excluded Expenses	Doyle, Lundholm and Soliman	2003/6	RAS	Intangibles
FailureProbability	Failure probability	Campbell, Hilscher and Szilagyi	2008/12	JF	Other
FailureProbabilityJune	Failure probability	Campbell, Hilscher and Szilagyi	2008/12	JF	Other
FEPS	Analyst earnings per share	Cen, Wei, and Zhang	2006/11	WP	Other
fgr5yrLag	Long-term EPS forecast	La Porta	1996/12	JF	Intangibles
fgr5yrNoLag	Long-term EPS forecast (Monthly)	La Porta	1996/12	JF	Intangibles
FirmAge	Firm age based on CRSP	Barry and Brown	1984/6	JFE	Other
FirmAgeMom	Firm Age - Momentum	Zhang	2004/7	JF	Momentum
ForecastDispersion	EPS Forecast Dispersion	Diether, Malloy and Scherbina	2002/10	JF	Intangibles
ForecastDispersionLT	Long-term forecast dispersion	Anderson, Ghysels, and Juergens	2005/9	RFS	Intangibles
FR	Pension Funding Status	Franzoni and Marin	2006/4	JF	Intangibles
FRbook	Pension Funding Status	Franzoni and Marin	2006/4	JF	Intangibles
Frontier	Efficient frontier index	Nguyen and Swanson	2009/2	JFQA	Intangibles
Governance	Governance Index	Gompers, Ishii and Metrick	2003/2	QJE	Other
GP	gross profits / total assets	Novy-Marx	2013/4	JFE	Profitability
GPlag	gross profits / total assets	Novy-Marx	2013/4	JFE	Profitability
GPlag-q	gross profits / total assets	Novy-Marx	2013/4	JFE	Profitability
GrAdExp	Growth in advertising expenses	Lou	2014/6	RFS	Intangibles
grcapx	Change in capex (two years)	Anderson and Garcia-Feijoo	2006/2	JF	Investment
grcapx1y	Investment growth (1 year)	Anderson and Garcia-Feijoo	2006/2	AR	Investment
grcapx3y	Change in capex (three years)	Anderson and Garcia-Feijoo	2006/2	JF	Investment
GrGMToGrSales	Gross margin growth to sales growth	Abarbanell and Bushee	1998/1	AR	Intangibles
GrLTNOA	Growth in long term operating assets	Fairfield, Whisenant and Yohn	2003/1	AR	Investment
GrSaleToGrInv	Sales growth over inventory growth	Abarbanell and Bushee	1998/1	AR	Intangibles
GrSaleToGrOverhead	Sales growth over overhead growth	Abarbanell and Bushee	1998/1	AR	Intangibles
GrSaleToGrReceivables	Change in sales vs change in receiv	Abarbanell and Bushee	1998/1	AR	Other
Herf	Industry concentration (sales)	Hou and Robinson	2006/8	JF	Intangibles
HerfAsset	Industry concentration (assets)	Hou and Robinson	2006/8	JF	Intangibles
HerfBE	Industry concentration (equity)	Hou and Robinson	2006/8	JF	Intangibles
High52	52 week high	George and Hwang	2004/10	JF	Momentum
hire	Employment growth	Bazdresch, Belo and Lin	2014/2	JPE	Intangibles
IdioRisk	Idiosyncratic risk	Ang et al.	2006/2	JF	Trading frictions
IdioVol3F	Idiosyncratic risk (3 factor)	Ang et al.	2006/2	JF	Trading frictions
IdioVolAHT	Idiosyncratic risk (AHT)	Ali, Hwang, and Trombley	2003/8	JFE	Trading frictions
IdioVolCAPM	Idiosyncratic risk (CAPM)	Ang et al.	2006/2	JF	Trading frictions
IdioVolQF	Idiosyncratic risk (q factor)	Ang et al.	2006/2	JF	Trading frictions
Illiquidity	Amihud's illiquidity	Amihud	2002/1	JFM	Trading frictions
IndIPO	Initial Public Offerings	Ritter	1991/3	JF	Intangibles
IndMom	Industry Momentum	Grinblatt and Moskowitz	1999/8	JFE	Momentum
IndRetBig	Industry return of big firms	Hou	2007/7	RFS	Momentum
IntanBM	Intangible return using BM	Daniel and Titman	2006/8	JF	Value vs. growth
IntanCFP	Intangible return using CFtoP	Daniel and Titman	2006/8	JF	Value vs. growth
IntanEP	Intangible return using EP	Daniel and Titman	2006/8	JF	Value vs. growth
IntanSP	Intangible return using Sale2P	Daniel and Titman	2006/8	JF	Value vs. growth
IntMom	Intermediate Momentum	Novy-Marx	2012/3	JFE	Momentum
IntrinsicValue	Intrinsic or historical value	Frankel and Lee	1998/6	JAE	Other
Investment	Investment to revenue	Titman, Wei and Xie	2004/12	JFQA	Investment
InvestPPEInv	change in ppe and inv/assets	Lyandres, Sun and Zhang	2008/11	RFS	Investment
InvGrowth	Inventory Growth	Belo and Lin	2012/1	RFS	Investment
IO-ShortInterest	Inst own among high short interest	Asquith Pathak and Ritter	2005/11	JFE	Other
iomom-cust	Customers momentum	Menzly and Ozbas	2010/8	JF	Momentum
iomom-supp	Suppliers momentum	Menzly and Ozbas	2010/8	JF	Momentum
KZ	Kaplan Zingales index	Lamont, Polk and Saa-Requejo	2001/4	RFS	Intangibles
KZ-q	Kaplan Zingales index quarterly	Lamont, Polk and Saa-Requejo	2001/4	RFS	Intangibles
LaborforceEfficiency	Laborforce efficiency	Abarbanell and Bushee	1998/1	AR	Other

Table A.1 – cont.

Acronym	Description	Author	Year/Month	Journal	Economic category
Leverage	Market leverage	Bhandari	1988/6	JFE	Profitability
Leverage-q	Market leverage quarterly	Bhandari	1988/6	JFE	Profitability
LRreversal	Long-run reversal	De Bondt and Thaler	1985/7	JF	Other
MaxRet	Maximum return over month	Bali, Cakici, and Whitelaw	2010/2	JF	Trading frictions
MeanRankRevGrowth	Revenue Growth Rank	Lakonishok, Shleifer, Vishny	1994/12	JF	Value vs. growth
Mom12m	Momentum (12 month)	Jegadeesh and Titman	1993/3	JF	Momentum
Mom12mOffSeason	Momentum without the seasonal part	Heston and Sadka	2008/2	JFE	Other
Mom6m	Momentum (6 month)	Jegadeesh and Titman	1993/3	JF	Momentum
Mom6mJunk	Junk Stock Momentum	Avramov et al	2007/10	JF	Momentum
MomOffSeason	Off season long-term reversal	Heston and Sadka	2008/2	JFE	Other
MomOffSeason06YrPlus	Off season reversal years 6 to 10	Heston and Sadka	2008/2	JFE	Other
MomOffSeason11YrPlus	Off season reversal years 11 to 15	Heston and Sadka	2008/2	JFE	Other
MomOffSeason16YrPlus	Off season reversal years 16 to 20	Heston and Sadka	2008/2	JFE	Other
MomRev	Momentum and LT Reversal	Chan and Ko	2006/1	JOIM	Momentum
MomSeason	Return seasonality years 2 to 5	Heston and Sadka	2008/2	JFE	Other
MomSeason06YrPlus	Return seasonality years 6 to 10	Heston and Sadka	2008/2	JFE	Other
MomSeason11YrPlus	Return seasonality years 11 to 15	Heston and Sadka	2008/2	JFE	Other
MomSeason16YrPlus	Return seasonality years 16 to 20	Heston and Sadka	2008/2	JFE	Other
MomSeasonShort	Return seasonality last year	Heston and Sadka	2008/2	JFE	Other
MomVol	Momentum in high volume stocks	Lee and Swaminathan	2000/10	JF	Momentum
MRreversal	Medium-run reversal	De Bondt and Thaler	1985/7	JF	Other
MS	Mohanram G-score	Mohanram	2005/9	RAS	Other
nanalyst	Number of analysts	Elgers, Lo and Pfeiffer	2001/10	AR	Other
NetDebtFinance	Net debt financing	Bradshaw, Richardson, Sloan	2006/10	JAE	Investment
NetDebtPrice	Net debt to price	Penman, Richardson and Tuna	2007/5	JAR	Value vs. growth
NetDebtPrice-q	Net debt to price	Penman, Richardson and Tuna	2007/5	JAR	Value vs. growth
NetEquityFinance	Net equity financing	Bradshaw, Richardson, Sloan	2006/10	JAE	Investment
NetPayoutYield	Net Payout Yield	Boudoukh et al.	2007/4	JF	Value vs. growth
NetPayoutYield-q	Net Payout Yield quarterly	Boudoukh et al.	2007/4	JF	Value vs. growth
NOA	Net Operating Assets	Hirshleifer et al.	2004/12	JAE	Investment
NumEarnIncrease	Earnings streak length	Loh and Warachka	2012/7	MS	Momentum
OperProf	operating profits / book equity	Fama and French	2006/12	JFE	Profitability
OperProfLag	operating profits / book equity	Fama and French	2006/12	JFE	Profitability
OperProfLag-q	operating profits / book equity	Fama and French	2006/12	JFE	Profitability
OperProfRD	Operating profitability RD adjusted	Ball et al.	2016/7	JFE	Profitability
OperProfRDLagAT	Oper prof RD adj lagged assets	Ball et al.	2016/7	JFE	Profitability
OperProfRDLagAT-q	Oper prof RD adj lagged assets (qtrly)	Ball et al.	2016/7	JFE	Profitability
OPLeverage	Operating leverage	Novy-Marx	2010/1	ROF	Intangibles
OPLeverage-q	Operating leverage (qtrly)	Novy-Marx	2010/1	ROF	Intangibles
OptionVolume1	Option to stock volume	Johnson and So	2012/11	JFE	Trading frictions
OptionVolume2	Option volume to average	Johnson and So	2012/11	JFE	Trading frictions
OrderBacklog	Order backlog	Rajgopal, Shevlin, Venkatachalam	2003/12	RAS	Intangibles
OrderBacklogChg	Change in order backlog	Baik and Ahn	2007/12	Other	Investment
OrgCap	Organizational capital	Eisfeldt and Papanikolaou	2013/8	JF	Intangibles
OrgCapNoAdj	Org cap w/o industry adjustment	Eisfeldt and Papanikolaou	2013/8	JF	Intangibles
OScore	O Score	Dichev	1998/6	JFE	Profitability
OScore-q	O Score quarterly	Dichev	1998/6	JFE	Profitability
PatentsRD	Patents to RD expenses	Hirschleifer, Hsu and Li	2013/3	JFE	Other
PayoutYield	Payout Yield	Boudoukh et al.	2007/4	JF	Value vs. growth
PayoutYield-q	Payout Yield quarterly	Boudoukh et al.	2007/4	JF	Value vs. growth
pchcurrat	Change in Current Ratio	Ou and Penman	1989/NA	JAR	Investment
pchdepr	Change in depreciation to PPE	Holthausen and Larcker	1992/6	JAE	Investment
pchgm-pchsale	Change in gross margin vs sales	Abarbanell and Bushee	1998/1	AR	Other
pchquick	Change in quick ratio	Ou and Penman	1989/NA	JAR	Investment
pchsaleinv	Change in sales to inventory	Ou and Penman	1989/NA	JAR	Other
PctAcc	Percent Operating Accruals	Hafzalla, Lundholm, Van Winkle	2011/1	AR	Investment
PctTotAcc	Percent Total Accruals	Hafzalla, Lundholm, Van Winkle	2011/1	AR	Investment
PM	Profit Margin	Soliman	2008/5	AR	Profitability
PM-q	Profit Margin	Soliman	2008/5	AR	Profitability
PredictedFE	Predicted Analyst forecast error	Frankel and Lee	1998/6	JAE	Intangibles
Price	Price	Blume and Husic	1972/5	JF	Other
PriceDelayRsq	Price delay r square	Hou and Moskowitz	2005/9	RFS	Trading frictions
PriceDelaySlope	Price delay coeff	Hou and Moskowitz	2005/9	RFS	Trading frictions
PriceDelayTstat	Price delay SE adjusted	Hou and Moskowitz	2005/9	RFS	Trading frictions
ProbInformedTrading	Probability of Informed Trading	Easley, Hvidkjaer and O'Hara	2002/10	JF	Trading frictions
PS	Piotroski F-score	Piotroski	2000/NA	AR	Other
PS-q	Piotroski F-score	Piotroski	2000/NA	AR	Other
quick	Quick ratio	Ou and Penman	1989/NA	JAR	Investment
RD	RD over market cap	Chan, Lakonishok and Sougiannis	2001/12	JF	Profitability
RD-q	RD over market cap quarterly	Chan, Lakonishok and Sougiannis	2001/12	JF	Profitability
rd-sale	RD to sales	Chan, Lakonishok and Sougiannis	2001/12	JF	Other
rd-sale-q	RD to sales	Chan, Lakonishok and Sougiannis	2001/12	JF	Other
RDAbility	RD ability	Cohen, Diether and Malloy	2013/3	RFS	Other
RDcap	RD capital-to-assets	Li	2011/9	RFS	Intangibles
RDIPO	IPO and no RD spending	Gou, Lev and Shi	2006/4	JBFA	Intangibles
RDS	Real dirty surplus	Landsman et al.	2011/1	AR	Intangibles
realestate	Real estate holdings	Tuzel	2010/6	RFS	Intangibles
ResidualMomentum	Momentum based on FF3 residuals	Blitz, Huij and Martens	2011/6	JEmpFin	Momentum
ResidualMomentum6m	6 month residual momentum	Blitz, Huij and Martens	2011/6	JEmpFin	Momentum
retConglomerate	Conglomerate return	Cohen and Lou	2012/5	JFE	Momentum
RetNOA	Return on Net Operating Assets	Soliman	2008/5	AR	Profitability
RetNOA-q	Return on Net Operating Assets	Soliman	2008/5	AR	Profitability
ReturnSkew	Return skewness	Bali, Engle and Murray	2015/4	Book	Trading frictions
ReturnSkew3F	Idiosyncratic skewness (3F model)	Bali, Engle and Murray	2015/4	Book	Trading frictions
ReturnSkewCAPM	Idiosyncratic skewness (CAPM)	Bali, Engle and Murray	2015/4	Book	Trading frictions
ReturnSkewQF	Idiosyncratic skewness (Q model)	Bali, Engle and Murray	2015/4	Book	Trading frictions
REV6	Earnings forecast revisions	Chan, Jegadeesh and Lakonishok	1996/12	JF	Momentum
RevenueSurprise	Revenue Surprise	Jegadeesh and Livnat	2006/4	JFE	Momentum
RIO-Disp	Inst Own and Forecast Dispersion	Nagel	2005/11	JF	Other
RIO-MB	Inst Own and Market to Book	Nagel	2005/11	JF	Other
RIO-Turnover	Inst Own and Turnover	Nagel	2005/11	JF	Other
RIO-Volatility	Inst Own and Idio Vol	Nagel	2005/11	JF	Other
roaq	Return on assets (qtrly)	Balakrishnan, Bartov and Faurel	2010/5	JAE	Profitability
roavol	RoA volatility	Francis, LaFond, Olsson, Schipper	2004/10	AR	Other
RoE	net income / book equity	Haugen and Baker	1996/7	JFE	Profitability
roic	Return on invested capital	Brown and Rowe	2007/6	WP	Profitability
salecash	Sales to cash ratio	Ou and Penman	1989/NA	JAR	Other
saleinv	Sales to inventory	Ou and Penman	1989/NA	JAR	Other
salerec	Sales to receivables	Ou and Penman	1989/NA	JAR	Other
secured	Secured debt	Valta	2016/2	JFQA	Intangibles
securedind	Secured debt indicator	Valta	2016/2	JFQA	Intangibles
sfe	Earnings Forecast to price	Elgers, Lo and Pfeiffer	2001/10	AR	Value vs. growth
sgr	Annual sales growth	Lakonishok, Shleifer, Vishny	1994/12	JF	Other
sgr-q	Annual sales growth quarterly	Lakonishok, Shleifer, Vishny	1994/12	JF	Other
ShareIss1Y	Share issuance (1 year)	Pontiff and Woodgate	2008/4	JF	Investment
ShareIss5Y	Share issuance (5 year)	Daniel and Titman	2006/8	JF	Investment
ShareRepurchase	Share repurchases	Ikenberry, Lakonishok, Vermaelen	1995/10	JFE	Investment
ShareVol	Share Volume	Datar, Naik and Radcliffe	1998/8	JFM	Trading frictions
ShortInterest	Short Interest	Dechow et al.	2001/7	JFE	Trading frictions
sinAlgo	Sin Stock (selection criteria)	Hong and Kacperczyk	2009/7	JFE	Other
Size	Size	Banz	1981/3	JFE	Other

Table A.1 – cont.

Acronym	Description	Author	Year/Month	Journal	Economic category
skew1	Volatility smirk near the money	Xing, Zhang and Zhao	2010/6	JFQA	Trading frictions
SmileSlope	Put volatility minus call volatility	Yan	2011/1	JFE	Trading frictions
SP	Sales-to-price	Barbee, Mukherji and Raines	1996/3	FAJ	Value vs. growth
SP-q	Sales-to-price quarterly	Barbee, Mukherji and Raines	1996/3	FAJ	Value vs. growth
Spinoff	Spinoffs	Cusatis, Miles and Woolridge	1993/6	JFE	Other
std-turn	Share turnover volatility	Chordia, Subra, Anshuman	2001/1	JFE	Trading frictions
STreversal	Short term reversal	Jegadeesh	1989/7	JF	Other
SurpriseRD	Unexpected RD increase	Eberhart, Maxwell and Siddique	2004/4	JF	Intangibles
tang	Tangibility	Hahn and Lee	2009/4	JF	Intangibles
tang-q	Tangibility quarterly	Hahn and Lee	2009/4	JF	Intangibles
Tax	Taxable income to income	Lev and Nissim	2004/10	AR	Profitability
Tax-q	Taxable income to income (qtrly)	Lev and Nissim	2004/10	AR	Profitability
TotalAccruals	Total accruals	Richardson et al.	2005/9	JAE	Investment
UpRecomm	Up Forecast	Barber et al.	2002/4	JF	Intangibles
VarCF	Cash-flow to price variance	Haugen and Baker	1996/7	JFE	Other
VolMkt	Volume to market equity	Haugen and Baker	1996/7	JFE	Trading frictions
VolSD	Volume Variance	Chordia, Subra, Anshuman	2001/1	JFE	Trading frictions
VolumeTrend	Volume Trend	Haugen and Baker	1996/7	JFE	Other
WW	Whited-Wu index	Whited and Wu	2006/6	RFS	Other
WW-Q	Whited-Wu index	Whited and Wu	2006/6	RFS	Other
XFIN	Net external financing	Bradshaw, Richardson, Sloan	2006/10	JAE	Investment
zerotrade	Days with zero trades	Liu	2006/12	JFE	Trading frictions
zerotradeAlt1	Days with zero trades	Liu	2006/12	JFE	Trading frictions
zerotradeAlt12	Days with zero trades	Liu	2006/12	JFE	Trading frictions
ZScore	Altman Z-Score	Dichev	1998/6	JFE	Profitability
ZScore-q	Altman Z-Score quarterly	Dichev	1998/6	JFE	Profitability

This table summarizes the firm characteristics used to construct the long-short anomalies. The columns show the acronym, a brief description, the original study, and the corresponding journal, where we follow Chen and Zimmermann (2022). In the column 'Economic category'we group similar factors based on their economic interpretation. Where available, we use the classification by Hou et al. (2020). For the remaining factors, we group them into the categories intangibles, investment, momentum, profitability, trading frictions, value vs. growth, and other.

Table A.2: Summary of Anomaly Variables Returns

Acronym	Economic category	R	t(R)	SD	SR	maxDD	Min	5%	95%	Max	Start	N
A. All anomalies												
Full sample		4.608	3.068	12.531	0.396	53.025	-20.574	-4.756	5.584	23.750	1957-03-20	760
Pre-publication		5.499	3.118	12.376	0.496	46.147	-18.150	-4.554	5.586	22.296	1957-03-20	558
Post-publication		2.558	0.869	12.113	0.208	38.226	-13.851	-5.052	5.494	13.005	2003-11-08	203
B. Grouped by econ. cat.												
	Intangibles	3.540	2.520	11.126	0.362	47.184	-17.730	-4.237	4.880	19.268	1969-07-06	606
	Investment	3.672	3.635	7.985	0.458	41.009	-10.587	-3.096	3.811	14.474	1957-07-20	762
	Momentum	8.907	5.839	15.194	0.739	56.200	-37.312	-5.553	6.557	21.159	1950-07-29	841
	Other	4.693	2.712	13.063	0.337	54.889	-19.024	-4.893	5.891	27.260	1956-02-09	771
	Profitability	5.028	2.819	13.450	0.381	57.680	-24.590	-5.183	5.790	21.285	1965-02-15	670
	Trading frictions	3.994	2.233	15.736	0.297	62.050	-23.198	-6.108	6.907	35.291	1942-10-13	933
	Value vs. growth	4.913	3.459	12.669	0.423	54.846	-22.937	-4.732	5.745	24.081	1955-12-27	780
C. Individual anomalies												
AbnormalAccruals	Investment	1.976	1.868	7.367	0.268	50.373	-5.307	-2.964	3.411	15.244	1972-07-31	582
AbnormalAccrualsPercent	Investment	-4.004	-7.250	3.866	-1.036	86.550	-5.818	-2.140	1.333	3.860	1972-01-31	588
AccrualQuality	Investment	0.702	0.388	14.464	0.049	65.655	-15.428	-5.990	6.701	31.358	1957-01-31	768
AccrualQualityJune	Investment	0.646	0.358	14.395	0.045	68.988	-14.969	-5.733	6.809	30.395	1957-07-31	762
Accruals	Investment	5.022	6.651	6.250	0.804	20.468	-8.761	-2.194	3.471	7.165	1952-07-31	822
AccrualsBM	Investment	14.974	6.299	17.854	0.839	41.987	-17.393	-5.982	8.985	31.215	1964-07-31	677
Activism1	Other	1.713	0.768	9.034	0.190	21.330	-8.014	-3.689	4.234	12.648	1990-10-31	197
Activism2	Intangibles	7.032	1.763	16.157	0.435	54.174	-12.828	-6.804	7.906	16.228	1990-10-31	197
AdExp	Intangibles	5.515	2.577	16.114	0.342	53.395	-20.097	-5.655	6.895	43.517	1955-07-29	680
AgeIPO	Intangibles	8.995	3.316	17.173	0.524	54.993	-23.749	-7.144	7.796	21.436	1980-12-31	481
AM	Value vs. growth	5.139	3.016	14.205	0.362	70.638	-28.135	-5.176	7.341	20.865	1951-07-31	834
AMq	Value vs. growth	8.071	3.587	16.912	0.477	74.096	-40.193	-5.875	8.328	21.328	1964-07-31	678
AnalystRevision	Momentum	7.801	8.734	5.981	1.304	24.575	-13.810	-2.132	3.265	5.551	1976-03-31	538
AnalystValue	Intangibles	2.143	1.044	13.697	0.156	67.921	-24.318	-5.662	5.507	23.206	1976-07-30	534
AnnouncementReturn	Momentum	13.184	14.300	6.481	2.034	13.483	-19.375	-1.533	3.363	6.943	1971-08-31	593
AOP	Intangibles	1.984	1.463	9.049	0.219	34.526	-9.994	-3.725	4.580	11.022	1976-07-30	534
AssetGrowth	Investment	10.953	7.360	12.317	0.889	35.141	-9.133	-3.978	6.394	28.474	1952-07-31	822
AssetGrowth_q	Investment	-7.125	-4.104	12.933	-0.551	99.140	-22.872	-7.349	4.458	12.747	1965-07-30	666
AssetLiquidityBook	Other	3.255	2.535	10.735	0.303	42.858	-13.004	-3.874	4.672	33.696	1951-02-28	839
AssetLiquidityBookQuart	Other	2.953	1.412	15.684	0.188	66.599	-19.197	-7.170	6.565	38.509	1964-10-30	675
AssetLiquidityMarket	Other	10.564	7.569	10.621	0.995	34.046	-15.988	-3.277	5.620	17.601	1963-02-28	695
AssetLiquidityMarketQuart	Other	8.732	5.001	13.115	0.666	47.901	-14.251	-4.816	6.710	19.520	1964-08-31	677
AssetTurnover	Other	4.373	4.310	8.397	0.521	41.306	-7.920	-3.595	4.324	10.475	1952-07-31	822
AssetTurnover_q	Other	6.451	4.680	10.453	0.617	31.334	-12.193	-4.277	5.167	16.280	1963-07-31	690
Beta	Trading frictions	3.987	1.436	26.753	0.149	91.975	-25.643	-10.518	12.171	66.200	1928-03-31	1,114
BetaBDLeverage	Trading frictions	3.587	1.969	12.555	0.286	49.749	-15.154	-5.209	6.087	17.786	1973-07-31	570
betaCC	Trading frictions	4.449	3.292	12.605	0.353	61.686	-11.579	-4.446	5.502	29.330	1934-01-31	1,044
betaCR	Trading frictions	-1.587	-1.925	7.686	-0.206	81.930	-16.951	-3.488	2.829	9.887	1934-01-31	1,044
BetaDimson	Trading frictions	-0.402	-0.266	14.673	-0.027	87.570	-20.169	-6.055	5.814	47.204	1926-09-30	1,132
BetaFP	Trading frictions	0.164	0.048	32.773	0.005	99.634	-27.002	-13.171	14.494	83.153	1929-02-28	1,103
BetaLiquidityPS	Trading frictions	3.365	2.111	11.723	0.287	45.425	-13.297	-4.829	5.695	14.063	1966-01-31	649
betaNet	Trading frictions	4.450	3.224	12.874	0.346	62.429	-12.213	-4.669	6.097	29.654	1934-01-31	1,044
betaRC	Trading frictions	-1.017	-0.639	14.838	-0.069	89.604	-36.464	-6.683	6.306	15.996	1934-01-31	1,044
betaRR	Trading frictions	1.869	0.831	20.971	0.089	90.344	-21.495	-8.518	9.390	55.245	1934-01-31	1,044
BetaSquared	Trading frictions	-3.494	-1.272	26.465	-0.132	99.935	-66.200	-12.089	10.425	25.817	1928-03-31	1,114
BetaTailRisk	Trading frictions	4.290	2.709	14.942	0.287	44.177	-19.120	-5.867	6.466	37.067	1932-01-30	1,068
betaVIX	Trading frictions	7.167	3.328	12.723	0.563	44.428	-13.073	-4.288	6.959	18.266	1986-02-28	419
BidAskSpread	Trading frictions	7.436	2.365	30.533	0.244	84.230	-22.807	-8.783	12.344	102.674	1926-09-30	1,132
BM	Value vs. growth	10.817	5.020	16.620	0.651	48.562	-25.557	-5.140	7.991	40.615	1961-07-31	714
BMdec	Value vs. growth	8.125	6.002	11.204	0.725	43.023	-16.354	-4.120	5.823	18.120	1952-07-31	822
BMq	Value vs. growth	11.959	4.936	18.212	0.657	52.478	-28.280	-5.351	8.163	39.226	1964-07-31	678
BookLeverage	Value vs. growth	1.559	1.203	10.806	0.144	61.967	-13.506	-4.096	4.229	28.613	1951-07-31	834
BookLeverageQuarterly	Value vs. growth	-0.832	-0.444	13.856	-0.060	75.590	-31.032	-5.928	6.171	15.706	1966-07-29	654
BPEBM	Value vs. growth	2.024	2.723	5.685	0.356	33.062	-7.161	-2.355	2.713	14.380	1962-07-31	702
BrandCapital	Intangibles	1.520	0.807	14.281	0.106	71.066	-21.931	-6.016	5.929	30.452	1955-07-29	690
BrandInvest	Intangibles	4.480	1.674	19.932	0.225	71.063	-24.679	-8.084	8.987	40.012	1965-07-30	666
CapTurnover	Other	2.629	2.368	9.156	0.287	49.854	-10.745	-4.058	4.341	10.646	1953-01-30	816
CapTurnover_q	Other	6.572	4.065	11.936	0.551	48.146	-17.783	-4.505	5.518	22.782	1966-07-29	654
Cash	Value vs. growth	7.705	3.137	17.236	0.447	62.280	-16.820	-6.668	7.818	46.972	1971-10-29	591
cashdebt	Other	-0.247	-0.140	14.696	-0.017	81.409	-33.187	-6.402	5.038	16.617	1952-01-31	828
CashProd	Intangibles	3.638	2.572	11.832	0.307	58.144	-27.924	-4.558	6.021	15.122	1951-01-31	840
CBOperProf	Profitability	5.936	3.523	12.886	0.461	53.472	-14.010	-5.300	6.505	17.411	1962-07-31	702
CBOperProfLagAT	Profitability	5.176	3.218	12.249	0.423	42.409	-27.702	-5.954	5.064	13.288	1963-01-31	696
CBOperProfLagAT_q	Profitability	9.091	5.741	11.151	0.815	39.919	-28.827	-4.442	4.766	14.235	1971-06-30	595
CF	Value vs. growth	4.474	2.562	14.556	0.307	50.586	-36.293	-5.488	5.886	17.254	1951-07-31	834
cfp	Value vs. growth	3.523	1.763	15.020	0.235	70.311	-32.368	-6.223	6.093	16.724	1964-07-31	678
cfpq	Value vs. growth	9.781	5.407	12.565	0.778	47.498	-29.978	-4.275	5.664	14.110	1972-10-31	579
CFq	Value vs. growth	13.434	6.362	16.264	0.826	72.984	-39.397	-5.160	6.670	32.551	1961-09-29	712
ChangeInRecommendation	Intangibles	6.745	7.128	4.924	1.370	7.211	-6.466	-1.408	2.642	6.657	1993-12-31	325
ChangeRoA	Profitability	11.065	7.787	10.393	1.065	38.746	-19.908	-2.886	4.576	17.454	1967-07-31	642
ChangeRoE	Profitability	10.813	6.858	11.533	0.938	32.843	-16.991	-3.180	4.249	47.698	1967-07-31	642
ChAssetTurnover	Profitability	1.940	3.857	4.132	0.470	16.633	-6.321	-1.663	1.933	5.648	1953-07-31	810
ChEQ	Intangibles	5.404	4.020	10.282	0.526	29.477	-14.126	-3.669	4.943	21.503	1962-07-31	702
ChForecastAccrual	Intangibles	2.607	3.914	4.445	0.587	10.817	-5.838	-1.744	2.224	5.006	1976-07-30	534
ChInv	Investment	7.200	8.028	7.423	0.970	24.401	-6.697	-2.644	4.178	15.967	1952-07-31	822
ChInvIA	Investment	4.207	6.234	5.586	0.753	29.420	-6.255	-1.936	2.868	10.375	1952-07-31	822
ChNAnalyst	Intangibles	12.088	1.840	37.447	0.323	99.981	-99.843	-10.287	13.574	55.667	1976-05-28	390
ChNCOA	Investment	-8.536	-8.978	7.898	-1.081	99.820	-15.440	-4.346	2.315	5.331	1952-01-31	828
ChNCOL	Investment	-4.400	-5.218	7.005	-0.628	97	-12.220	-3.453	2.570	5.754	1952-01-31	828
ChNNCOA	Investment	2.716	5.350	4.203	0.646	13.783	-4.704	-1.723	2.279	5.299	1952-07-31	822
ChNWC	Profitability	1.814	4.300	3.492	0.520	24.108	-4.351	-1.324	1.851	6.097	1952-07-31	822
ChPM	Other	1.709	3.067	4.613	0.371	19.056	-4.448	-1.968	2.373	4.668	1952-07-31	822

Table A.2 – cont.

Acronym	Economic category	R	t(R)	SD	SR	maxDD	Min	5%	95%	Max	Start	N
ChTax	Intangibles	11.678	10.049	8.870	1.317	21.738	-20.135	-2.648	4.727	18.141	1962-10-31	699
CitationsRD	Other	2.369	2.182	6.332	0.374	32.343	-8.176	-2.743	3.063	9.089	1977-07-29	408
CompEquIss	Investment	3.805	3.068	11.763	0.323	57.380	-19.376	-3.639	3.766	42.241	1931-01-31	1,080
CompositeDebtIssuance	Investment	2.813	5.327	4.241	0.663	16.535	-5.517	-1.810	2.180	3.866	1956-07-31	774
ConsRecomm	Other	5.891	2.409	12.746	0.462	42.565	-12.732	-5.186	6.761	19.279	1993-11-30	326
ConvDebt	Intangibles	2.831	4.260	5.561	0.509	20.422	-23.012	-1.979	2.489	5.700	1951-01-31	840
CoskewACX	Trading frictions	4.469	3.562	9.360	0.477	35.927	-11.753	-3.452	4.608	19.586	1963-07-31	668
Coskewness	Trading frictions	1.112	1.153	9.348	0.119	70.662	-21.274	-3.576	4.241	14.991	1927-01-31	1,128
CredRatDG	Profitability	8.212	3.145	14.537	0.565	56.133	-23.704	-6.132	6.529	12.815	1986-02-28	372
currat	Value vs. growth	2.300	2.050	9.385	0.245	50.149	-14.905	-3.680	4.236	26.140	1951-01-31	840
CustomerMomentum	Other	7.437	2.136	22.965	0.324	87.213	-61.346	-6.128	9.473	36.098	1977-07-29	522
DebtIssuance	Investment	3.577	5.702	4.391	0.815	13.197	-4.961	-1.609	2.145	9.098	1972-01-31	588
DelayAcct	Other	-1.616	-0.971	10.809	-0.150	69.412	-15.681	-4.882	4.873	12.614	1978-07-31	506
DelayNonAcct	Other	0.936	0.696	8.732	0.107	38.014	-9.843	-4.080	4.032	10.909	1978-07-31	506
DelBreadth	Intangibles	6.880	2.899	14.976	0.459	42.848	-29.964	-5.476	6.541	29.975	1980-07-31	478
DelCOA	Investment	4.658	6.201	6.217	0.749	31.089	-6.524	-2.259	3.439	10.706	1952-07-31	822
DelCOL	Investment	2.536	3.499	5.998	0.423	32.895	-6.617	-2.508	3.325	6.655	1952-07-31	822
DelDRC	Profitability	8.271	1.582	23.673	0.349	48.281	-42.979	-3.801	5.549	42.596	2000-07-31	246
DelEqu	Investment	5.585	3.969	10.761	0.519	36.342	-8.165	-3.651	5.597	21.525	1962-07-31	702
DelFINL	Investment	6.152	11.461	4.443	1.385	18.264	-5.483	-1.545	2.520	7.868	1952-07-31	822
DelLTI	Investment	1.935	3.605	4.409	0.439	18.242	-4.580	-1.613	1.942	9.633	1953-07-31	810
DelNetFin	Investment	4.046	6.924	4.837	0.837	21.473	-7.375	-1.840	2.489	4.947	1952-07-31	822
DelSTI	Investment	0.542	0.848	4.538	0.119	40.103	-3.884	-1.682	2.164	8.190	1970-07-31	606
depr	Other	5.431	3.677	12.357	0.440	51.003	-14.443	-4.405	5.822	30.729	1951-01-31	840
DivInit	Value vs. growth	4.128	3.121	12.846	0.321	49.172	-43.755	-3.722	5.071	34.573	1926-09-30	1,132
DivOmit	Value vs. growth	7.790	4.326	17.168	0.454	62.135	-26.050	-6.891	7.933	42.036	1927-11-30	1,091
DivSeason	Value vs. growth	3.675	14.457	2.463	1.492	7.132	-3.414	-0.814	1.420	4.680	1927-02-28	1,127
DivYield	Value vs. growth	6.906	3.218	20.847	0.331	77.561	-31.775	-8.075	9.405	88.791	1926-09-30	1,132
DivYieldAnn	Value vs. growth	-1.355	-1.365	9.638	-0.141	91.044	-35.937	-3.324	3.137	7.769	1926-09-30	1,132
DivYieldST	Value vs. growth	6.481	9.101	6.913	0.937	17.290	-11.302	-1.973	3.229	34.660	1926-09-30	1,131
dNoa	Investment	9.251	8.419	8.404	1.101	17.888	-14.126	-2.654	4.623	16.915	1962-07-31	702
DolVol	Trading frictions	8.803	4.040	21.162	0.416	46.791	-16.811	-6.091	7.872	83.559	1926-09-30	1,132
DownRecomm	Intangibles	4.477	6.100	3.819	1.172	5.991	-5.991	-1.061	2.077	7.208	1993-12-31	325
DownsideBeta	Trading frictions	1.234	0.579	20.712	0.060	81.963	-35.682	-9.199	9.017	36.892	1926-09-30	1,132
EarningsConservatism	Other	-0.046	-0.091	3.945	-0.012	24.520	-7.691	-1.775	1.762	4.180	1960-01-29	732
EarningsConsistency	Intangibles	3.118	3.803	6.735	0.463	36.496	-8.200	-3.047	3.239	6.818	1953-07-31	810
EarningsForecastDisparity	Intangibles	5.295	3.518	9.399	0.563	35.356	-14.629	-3.599	4.209	10.182	1982-01-29	468
EarningsPersistence	Other	-1.665	-1.688	7.641	-0.218	76.584	-11.517	-3.704	3.353	8.554	1961-01-31	720
EarningsPredictability	Other	-6.669	-4.644	11.123	-0.600	99.146	-10.177	-5.641	4.293	20.682	1961-01-31	720
EarningsSmoothness	Other	1.915	1.442	10.283	0.186	44.334	-10.875	-4.432	4.712	18.510	1961-01-31	720
EarningsStreak	Other	10.655	10.064	6.367	1.673	14.726	-14.726	-1.678	3.672	5.792	1984-11-30	434
EarningsSurprise	Momentum	8.607	9.543	6.844	1.258	32.039	-13.240	-2.326	3.658	9.629	1963-06-28	691
EarningsTimeliness	Other	0.559	0.904	4.823	0.116	36.643	-6.167	-1.969	2.269	10.844	1960-01-29	732
EarningsValueRelevance	Other	0.683	1.154	4.583	0.149	29.746	-6.322	-2.007	2.089	4.853	1961-01-31	720
EarnSupBig	Momentum	3.925	3.082	9.664	0.406	32.411	-17.367	-4.074	4.276	12.992	1963-06-28	691
EBM	Value vs. growth	2.546	3.505	5.556	0.458	23.888	-5.635	-2.281	2.856	10.374	1962-07-31	702
EBM_q	Value vs. growth	7.464	5.685	9.603	0.777	28.163	-17.715	-3.055	4.308	21.200	1967-04-28	642
EntMult	Value vs. growth	7.700	5.418	11.848	0.650	49.869	-17.387	-4.551	6.280	19.392	1951-07-31	834
EntMult_q	Value vs. growth	-16.156	-9.121	12.955	-1.247	99.994	-30.493	-6.224	3.222	18.873	1967-04-28	642
EP	Value vs. growth	3.393	2.909	9.760	0.348	41.611	-15.944	-3.879	4.803	17.139	1951-01-31	840
EPq	Value vs. growth	13.652	12.360	8.472	1.611	37.400	-13.883	-2.369	4.906	12.790	1962-03-30	706
EquityDuration	Value vs. growth	4.893	2.578	14.394	0.340	66.695	-16.086	-6.031	7.160	20.939	1963-07-31	690
ETR	Other	-0.609	-0.800	5.908	-0.103	52.348	-13.621	-2.026	1.763	11.243	1960-07-29	723
ExchSwitch	Trading frictions	7.486	4.924	11.596	0.646	26.117	-13.520	-4.582	6.374	12.930	1962-11-30	698
ExclExp	Intangibles	1.438	1.049	8.307	0.173	20.432	-34.727	-2.163	2.663	7.241	1983-07-29	440
FailureProbability	Other	2.237	0.756	21.551	0.104	82.379	-41.921	-9.200	8.860	30.412	1968-01-31	636
FailureProbabilityJune	Other	-0.584	-0.204	20.970	-0.028	91.037	-19.760	-8.391	9.824	45.198	1967-07-31	642
FEPS	Other	8.843	2.848	20.811	0.425	80.805	-31.770	-7.216	10.261	31.558	1976-02-27	539
fgr5yrLag	Intangibles	1.847	0.646	17.732	0.104	68.426	-27.447	-6.775	7.820	18.003	1982-07-30	462
fgr5yrNoLag	Intangibles	-2.901	-0.955	18.958	-0.153	88.586	-25.425	-8.817	7.020	28.928	1982-01-29	468
FirmAge	Other	-0.681	-0.839	7.807	-0.087	71.189	-17.350	-3.511	3.355	16.920	1928-07-31	1,110
FirmAgeMom	Momentum	14.884	7.370	19.397	0.767	85.360	-43.680	-7.073	8.982	32.022	1926-12-31	1,107
ForecastDispersion	Intangibles	5.198	2.273	15.323	0.339	59.378	-25.571	-6.720	6.970	13.746	1976-02-27	539
ForecastDispersionLT	Intangibles	-0.210	-0.125	10.478	-0.020	64.888	-16.820	-4.310	3.542	17.539	1982-01-29	468
FR	Intangibles	-0.760	-0.397	12.033	-0.063	73.811	-20.101	-5.593	4.640	13.236	1981-07-31	474
FRbook	Intangibles	0.727	0.667	6.882	0.106	47.642	-10.080	-3.200	3.085	7.885	1981-02-27	479
Frontier	Intangibles	15.599	6.651	17.785	0.877	41.978	-18.694	-6.052	9.343	25.927	1963-07-31	690
Governance	Other	-1.528	-0.583	10.627	-0.144	62.263	-11.517	-4.734	4.404	12.113	1990-10-31	197
GP	Profitability	4.333	3.549	10.178	0.426	55.799	-15.813	-4.124	5.190	12.305	1951-07-31	834
GPlag	Profitability	2.761	2.622	8.747	0.316	36.922	-8.688	-4.167	4.034	11.789	1952-01-31	828
GPlag_q	Profitability	8.400	4.878	12.054	0.697	56.507	-22.770	-4.786	5.319	10.972	1972-01-31	588
GrAdExp	Intangibles	2.849	1.503	14.123	0.202	80.327	-25.982	-4.973	5.514	24.665	1965-07-30	666
grcapx	Investment	4.065	5.625	5.937	0.685	22.295	-5.778	-2.317	3.218	8.255	1953-07-31	810
grcapx1y	Investment	-1.875	-2.925	5.285	-0.355	83.330	-7.927	-2.415	2.054	13.186	1953-01-30	816
grcapx3y	Investment	4.450	5.477	6.626	0.672	23.556	-6.536	-2.484	3.636	13.852	1954-07-30	798
GrGMToGrSales	Intangibles	2.443	3.806	5.312	0.460	19.093	-5.611	-2.385	2.712	5.316	1952-07-31	822
GrLTNOA	Investment	2.380	3.168	6.218	0.383	21.305	-8.990	-2.770	3.137	8.077	1952-07-31	822
GrSaleToGrInv	Intangibles	3.082	5.449	4.681	0.658	19.914	-4.589	-2.008	2.263	6.394	1952-07-31	822
GrSaleToGrOverhead	Intangibles	-0.104	-0.152	5.696	-0.018	50.442	-12.398	-2.472	2.327	9.850	1952-07-31	822
GrSaleToGrReceivables	Other	1.481	2.945	4.163	0.356	21.693	-4.096	-1.686	2.015	5.631	1952-07-31	822
Herf	Intangibles	1.188	1.508	6.594	0.180	51.465	-7.384	-2.490	2.819	18.758	1951-01-31	840
HerfAsset	Intangibles	0.454	0.497	7.584	0.060	61.889	-11.207	-2.654	3.121	30.242	1951-12-31	829
HerfBE	Intangibles	1.026	1.138	7.495	0.137	52.971	-11.185	-2.725	2.831	28.812	1951-12-31	829
High52	Momentum	0.261	0.107	23.702	0.011	99.701	-69.538	-9.684	7.318	18.550	1926-09-30	1,132
hire	Intangibles	5.168	5.322	7.234	0.714	27.421	-9.683	-2.733	4.139	11.145	1965-07-30	666
IdioRisk	Trading frictions	6.499	2.830	22.303	0.291	88.132	-39.119	-9.665	9.549	38.075	1926-09-30	1,132
IdioVol3F	Trading frictions	5.416	2.322	22.651	0.239	91.773	-43.016	-9.711	9.366	39.316	1926-09-30	1,132
IdioVolAHT	Trading frictions	2.220	0.906	23.701	0.094	96.593	-43.551	-11.245	9.853	35.077	1927-07-30	1,122

Table A.2 – cont.

Acronym	Economic category	R	t(R)	SD	SR	maxDD	Min	5%	95%	Max	Start	N
IdioVolCAPM	Trading frictions	0.909	0.329	26.816	0.034	99.137	-26.001	-9.538	11.239	82.801	1926-09-30	1,132
IdioVolQF	Trading frictions	-3.189	-0.980	23.695	-0.135	97.752	-25.702	-9.634	10.463	52.355	1967-02-28	636
Illiquidity	Trading frictions	4.134	3.344	11.956	0.346	53.608	-11.582	-4.836	6.127	33.749	1927-07-30	1,122
IndIPO	Intangibles	4.646	2.673	11.743	0.396	37.676	-18.905	-4.542	5.156	15.404	1975-05-30	548
IndMom	Momentum	4.559	4	11.071	0.412	57.470	-27.742	-3.733	4.654	25.526	1926-09-30	1,132
IndRetBig	Momentum	16.430	11.842	13.475	1.219	64.652	-24.055	-4.225	7.065	34.887	1926-09-30	1,132
IntanBM	Value vs. growth	2.913	1.705	12.614	0.231	44.075	-18.729	-4.872	6.138	24.469	1966-07-29	654
IntanCFP	Value vs. growth	3.585	2.499	11.567	0.310	44.670	-27.887	-4.267	5.622	24.293	1956-01-31	780
IntanEP	Value vs. growth	3.242	2.781	9.365	0.346	40.047	-13.871	-3.601	4.412	16.625	1956-07-31	774
IntanSP	Value vs. growth	4.624	2.589	14.347	0.322	62.892	-13.204	-5.376	6.803	23.346	1956-07-31	774
IntMom	Momentum	13.722	5.808	22.907	0.599	88.552	-83.162	-8.604	10.899	20.057	1927-01-31	1,128
IntrinsicValue	Other	3.068	1.372	14.915	0.206	60.048	-24.942	-5.703	6.730	17.221	1976-07-30	534
Investment	Investment	2.108	1.918	9.030	0.233	54.311	-12.673	-3.677	4.048	26.882	1953-07-31	810
InvestPPEInv	Investment	6.598	8.666	6.324	1.043	28.493	-7.032	-2.053	3.457	13.252	1952-01-31	828
InvGrowth	Investment	7.771	7.191	8.945	0.869	34.881	-8.768	-3.457	4.663	16.469	1952-07-31	822
IO_ShortInterest	Other	34.320	4.803	45.799	0.749	70.314	-58.515	-17.895	24.400	46.499	1979-11-30	493
iomom_cust	Momentum	7.264	3.175	13.518	0.537	44.183	-33.240	-4.821	6.553	15.782	1986-02-28	419
iomom_supp	Momentum	7.028	2.953	14.044	0.500	39.746	-22.748	-5.279	6.547	20.423	1986-02-28	418
KZ	Intangibles	0.843	0.618	10.444	0.081	63.554	-17.243	-4.576	4.615	13.567	1962-07-31	702
KZ_q	Intangibles	-9.538	-3.017	20.075	-0.475	99.661	-36.364	-6.295	4.441	68.723	1972-04-28	484
LaborforceEfficiency	Other	-0.013	-0.025	4.346	-0.003	44.112	-5.156	-2.043	2.087	5.006	1952-07-31	822
Leverage	Profitability	3.853	2.299	13.969	0.276	75.307	-33.085	-4.922	7.008	20.942	1951-07-31	834
Leverage_q	Profitability	5.283	2.342	16.651	0.317	77.886	-41.682	-5.998	7.953	17.376	1966-07-29	654
LRreversal	Other	7.813	3.621	20.696	0.378	67.396	-22.419	-5.614	7.665	75.696	1929-01-31	1,104
MaxRet	Trading frictions	7.305	2.792	25.414	0.287	84.963	-45.188	-10.588	11.060	48.034	1926-09-30	1,132
MeanRankRevGrowth	Value vs. growth	2.700	3.063	7.026	0.384	33.692	-7.794	-2.818	3.491	9.186	1957-07-31	762
Mom12m	Momentum	9.858	3.393	28.164	0.350	99.532	-88.699	-11.343	10.371	29.484	1927-01-31	1,128
Mom12mOffSeason	Other	10.356	3.851	26.118	0.397	97.148	-87.633	-9.998	9.761	29.996	1926-09-30	1,132
Mom6m	Momentum	7.017	2.625	25.960	0.270	99.264	-77.393	-9.543	8.804	32.091	1926-09-30	1,132
Mom6mJunk	Momentum	11.686	3.346	21.627	0.540	53.689	-36.607	-7.183	8.839	42.169	1978-12-29	460
MomOffSeason	Other	11.778	5.648	20.118	0.585	63.137	-14.896	-5.957	9.280	59.863	1927-12-31	1,117
MomOffSeason06YrPlus	Other	7.854	6.034	12.286	0.639	43.147	-31.543	-3.781	5.546	41.583	1931-12-31	1,069
MomOffSeason11YrPlus	Other	3.205	2.773	10.598	0.302	27.372	-9.702	-3.796	4.190	51.967	1936-12-31	1,009
MomOffSeason16YrPlus	Other	3.553	3.358	9.290	0.382	40.307	-9.647	-3.904	4.107	22.418	1943-12-31	925
MomRev	Momentum	7.607	3.593	20.297	0.375	95.380	-60.502	-7.580	8.646	35.977	1929-01-31	1,103
MomSeason	Other	9.042	6.786	12.851	0.704	58.009	-27.835	-4.145	6.272	24.178	1928-01-31	1,116
MomSeason06YrPlus	Other	8.014	7.009	10.786	0.743	39.311	-21.163	-3.203	5.274	32.475	1932-01-30	1,068
MomSeason11YrPlus	Other	6.435	7.117	8.287	0.777	23.154	-18.298	-2.924	4.337	12.701	1937-01-30	1,008
MomSeason16YrPlus	Other	6.026	6.577	8.144	0.740	16.833	-8.477	-2.890	3.969	17.795	1942-01-31	948
MomSeasonShort	Other	10.333	6.261	16.001	0.646	76.656	-54.927	-5.432	6.589	43.246	1927-01-31	1,128
MomVol	Momentum	11.596	3.796	29.459	0.394	99.313	-68.026	-10.975	11.548	37.009	1928-01-31	1,116
MRreversal	Other	4.957	3.162	15.159	0.327	67.114	-13.021	-4.232	5.419	61.602	1927-07-30	1,122
MS	Other	12.290	5.003	16.464	0.746	32.814	-19.743	-4.647	7.197	65.859	1974-09-30	539
nanalyst	Other	-0.634	-0.483	8.795	-0.072	58.317	-14.030	-4.092	3.846	10.109	1976-02-27	539
NetDebtFinance	Investment	7.729	8.854	6.079	1.271	15.284	-4.989	-2.139	3.568	8.317	1972-07-31	582
NetDebtPrice	Value vs. growth	6.727	4.133	12.343	0.545	35.617	-15.666	-4.773	5.217	21.375	1963-07-31	690
NetDebtPrice_q	Value vs. growth	-9.010	-4.434	14.371	-0.627	99.695	-21.181	-6.650	4.800	23.187	1970-11-30	600
NetEquityFinance	Investment	10.836	5.378	14.033	0.772	53.302	-26.395	-5.692	6.355	18.058	1972-07-31	582
NetPayoutYield	Value vs. growth	7.672	4.332	14.551	0.527	59.081	-23.302	-5.789	7.424	17.538	1953-07-31	810
NetPayoutYield_q	Value vs. growth	5.264	2.148	18.053	0.292	64.716	-30.815	-7.553	8.439	36.091	1966-10-31	651
NOA	Investment	10.314	7.101	11.062	0.932	40.426	-12.749	-3.967	5.737	27.470	1963-01-31	696
NumEarnIncrease	Momentum	5.225	9.206	4.286	1.219	19.702	-6.137	-1.620	2.415	5.178	1964-01-31	684
OperProf	Profitability	4.876	3.143	11.765	0.414	54.613	-25.482	-4.121	4.563	19.835	1963-07-31	690
OperProfLag	Profitability	2.222	1.491	11.251	0.197	55.295	-27.828	-4.454	4.514	19.342	1964-01-31	684
OperProfLag_q	Profitability	6.731	3.001	16.404	0.410	68.712	-39.694	-6.718	6.686	18.946	1967-04-28	642
OperProfRD	Profitability	4.599	2.311	15.091	0.305	66.236	-14.319	-6.737	7.788	17.573	1963-07-31	690
OperProfRDLagAT	Profitability	1.189	0.756	13.065	0.091	72.754	-28.402	-6.199	4.781	12.783	1952-01-31	828
OperProfRDLagAT_q	Profitability	10.990	4.371	17.601	0.624	67.867	-45.839	-7.098	7.137	19.732	1972-01-31	588
OPLeverage	Intangibles	4.744	3.818	10.357	0.458	39.750	-12.354	-4.142	5.035	20.604	1951-07-31	834
OPLeverage_q	Intangibles	5.831	3.274	13.118	0.445	42.393	-15.361	-5.231	6.544	21.029	1966-07-29	651
OptionVolume1	Trading frictions	5.240	2.021	12.920	0.406	55.349	-23.404	-4.069	5.505	19.543	1996-03-29	298
OptionVolume2	Trading frictions	3.847	1.754	10.913	0.353	24.647	-9.058	-2.761	2.686	36.023	1996-04-30	297
OrderBacklog	Intangibles	0.715	0.601	8.369	0.085	53.035	-9.697	-3.716	3.514	15.790	1971-07-30	594
OrderBacklogChg	Investment	4.194	2.567	11.378	0.369	44.392	-18.400	-4.450	5.326	13.777	1972-07-31	582
OrgCap	Intangibles	4.573	4.160	9.165	0.499	33.984	-10.615	-3.832	4.604	11.703	1951-07-31	834
OrgCapNoAdj	Intangibles	7.670	4.989	12.818	0.598	48.007	-17.128	-4.183	6.163	33.081	1951-07-31	834
OScore	Profitability	8.459	3.276	18.076	0.468	65.410	-42.947	-7.634	8.330	20.511	1972-01-31	588
OScore_q	Profitability	-10.965	-3.899	17.106	-0.641	99.422	-13.951	-7.450	8.042	22.967	1984-01-31	444
PatentsRD	Other	2.957	2.921	5.637	0.525	20.997	-4.651	-2.093	2.581	10.977	1977-07-29	372
PayoutYield	Value vs. growth	2.535	2.044	10.188	0.249	55.369	-9.169	-4.373	5.087	16.252	1953-07-31	810
PayoutYield_q	Value vs. growth	5.439	4.427	9.049	0.601	30.056	-13.113	-3.384	4.117	16.620	1966-10-31	651
pchcurrat	Investment	0.109	0.192	4.697	0.023	59.192	-11.203	-2.183	1.840	3.711	1952-07-31	822
pchdepr	Investment	1.918	3.133	5.087	0.377	23.974	-5.578	-1.833	2.331	11.422	1952-01-31	828
pchgm_pchsale	Other	2.654	4.191	5.260	0.505	17.225	-7.684	-2.483	2.545	4.860	1952-01-31	828
pchquick	Investment	1.021	1.752	4.825	0.212	56.445	-11.420	-2.143	2.045	3.456	1952-07-31	822
pchsaleinv	Other	4.176	7.785	4.456	0.937	32.391	-3.709	-1.716	2.496	5.249	1952-01-31	828
PctAcc	Investment	4.770	4.612	7.773	0.614	22.690	-8.329	-2.878	4.172	12.633	1964-07-31	678
PctTotAcc	Investment	3.848	4.090	5.364	0.717	20.510	-7.088	-1.952	2.906	5.433	1988-07-29	390
PM	Profitability	-0.338	-0.264	10.657	-0.032	77.029	-22.804	-4.291	4.816	16.950	1951-07-31	834
PM_q	Profitability	7.025	2.947	18.363	0.383	66.098	-47.176	-8.132	7.529	20.543	1961-09-29	712
PredictedFE	Intangibles	0.009	0.004	13.560	0.001	71.065	-21.731	-6.157	6.248	13.841	1983-07-29	450
Price	Other	9.250	3.029	29.665	0.312	79.409	-24.709	-8.117	12.876	100.833	1926-09-30	1,132
PriceDelayRsq	Trading frictions	6.113	3.327	17.729	0.345	43.594	-31.971	-5.755	7.314	64.449	1927-08-31	1,117
PriceDelaySlope	Trading frictions	2.421	2.460	9.496	0.255	25.608	-11.884	-3.785	4.076	29.902	1927-08-31	1,117
PriceDelayTstat	Trading frictions	0.308	0.392	7.584	0.041	60.189	-15.351	-3.429	3.129	17.205	1927-08-31	1,117
ProbInformedTrading	Trading frictions	16.034	4.325	16.159	0.992	27.940	-25.799	-5.687	7.995	15.767	1984-02-29	228
PS	Other	9.520	2.824	23.601	0.403	61.673	-37.699	-8.800	10.718	38.949	1972-01-31	588
PS_q	Other	10.286	5.751	10.880	0.945	47.416	-13.940	-4.492	5.216	12.435	1984-01-31	444

Table A.2 – cont.

Acronym	Economic category	R	t(R)	SD	SR	maxDD	Min	5%	95%	Max	Start	N
quick	Investment	2.651	2.137	10.377	0.255	61.508	-17.187	-4.158	4.341	29.082	1951-01-31	840
RD	Profitability	10.822	5.567	16.206	0.668	45.119	-15.353	-5.082	8.037	50.778	1951-07-31	834
RD_q	Profitability	17.385	4.260	23.083	0.753	36.337	-14.516	-6.727	12.271	45.744	1989-01-31	384
rd_sale	Other	2.112	0.923	18.933	0.112	85.143	-18.465	-7.065	7.436	61.738	1952-07-31	822
rd_sale_q	Other	2.777	0.672	23.002	0.121	85.224	-17.849	-8.758	9.696	58.856	1990-01-31	372
RDAbility	Other	0.127	0.079	12.805	0.010	73.118	-12.956	-6.043	6.068	14.978	1957-07-31	762
RDcap	Intangibles	5.563	2.954	11.985	0.464	39.292	-9.083	-4.353	6.098	20.935	1980-07-31	486
RDIPO	Intangibles	7.986	3.291	16.034	0.498	54.932	-26.903	-6	7.171	19.231	1977-01-31	524
RDS	Intangibles	3.099	2.550	8.377	0.370	29.631	-14.937	-2.957	4.063	11.682	1973-07-31	570
realestate	Intangibles	3.246	2.257	10.221	0.318	36.575	-15.268	-4.227	4.959	11.786	1970-07-31	606
ResidualMomentum	Momentum	10.279	8.113	12.058	0.852	43.496	-29.360	-4.184	5.683	17.812	1930-06-30	1,087
ResidualMomentum6m	Momentum	4.250	4.001	10.132	0.419	37.327	-23.253	-4.004	4.079	13.009	1930-01-31	1,092
retConglomerate	Momentum	13.997	6.753	13.433	1.042	24.048	-16.426	-4.368	8.037	21.581	1976-02-27	504
RetNOA	Profitability	0.141	0.152	7.065	0.020	47.799	-10.833	-2.981	2.833	17.562	1963-07-31	690
RetNOA_q	Profitability	6.997	3.255	16.123	0.434	63.183	-36.101	-6.855	7.057	18.418	1964-10-30	675
ReturnSkew	Trading frictions	5.803	7.819	7.208	0.805	31.453	-18.157	-2.110	3.251	12.179	1926-09-30	1,132
ReturnSkew3F	Trading frictions	4.497	7.882	5.542	0.812	26.474	-13.135	-1.607	2.652	10.253	1926-09-30	1,132
ReturnSkewCAPM	Trading frictions	-4.917	-7.208	6.626	-0.742	99.308	-11.281	-2.770	1.868	21.234	1926-09-30	1,132
ReturnSkewQF	Trading frictions	-2.863	-4.357	4.784	-0.598	80.491	-8.962	-2.293	1.520	10.119	1967-02-28	636
REV6	Momentum	9.534	4.131	15.368	0.620	64.114	-34.365	-6.128	6.317	14.066	1976-09-30	532
RevenueSurprise	Momentum	7.233	8.584	6.394	1.131	18.352	-12.137	-1.746	2.927	14.765	1963-06-28	691
RIO_Disp	Other	7.766	3.534	14.688	0.529	52.278	-16.378	-5.047	7.493	25.716	1976-02-27	536
RIO_MB	Other	8.370	4.166	15.259	0.549	70.199	-19.430	-5.652	7.779	26.724	1963-01-31	692
RIO_Turnover	Other	3.851	2.481	15.063	0.256	81.039	-20.468	-6.690	7.233	18.778	1926-09-30	1,130
RIO_Volatility	Other	5.680	3.061	18	0.316	87.004	-21.933	-7.469	8.181	50.584	1926-09-30	1,129
roaq	Profitability	13.804	5.206	19.575	0.705	66.879	-33.620	-7.489	8.434	42.393	1966-07-29	654
roavol	Other	0.892	0.311	20.792	0.043	86.569	-21.953	-8.471	7.880	39.757	1968-06-28	631
RoE	Profitability	2.745	2.319	9.130	0.301	48.518	-22.086	-3.388	4.249	14.631	1961-07-31	714
roic	Profitability	0.333	0.161	15.832	0.021	75.953	-36.024	-7.025	5.713	18.188	1962-07-31	702
salecash	Other	0.828	0.710	9.756	0.085	63.509	-25.100	-4.047	3.899	14.508	1951-01-31	840
saleinv	Other	2.481	2.989	6.943	0.357	28.359	-11.825	-3.104	3.137	6.522	1951-01-31	840
salerec	Other	2.156	2.573	7.012	0.308	44.771	-6.142	-2.974	3.257	11.471	1951-01-31	840
secured	Intangibles	-0.789	-0.859	5.702	-0.138	47.464	-7.671	-2.361	2.406	7.322	1982-07-30	462
securedind	Intangibles	-0.052	-0.054	6.096	-0.009	51.711	-7.257	-2.212	2.146	13.593	1981-01-30	480
sfe	Value vs. growth	5.529	1.714	21.573	0.256	88.500	-54.648	-8.490	9.102	21.883	1976-04-30	537
sgr	Other	-5.431	-5.847	7.716	-0.704	98.533	-14.800	-4.258	2.937	7.272	1952-01-31	828
sgr_q	Other	4.117	3.150	9.980	0.412	36.258	-20.083	-4.513	4.033	12.006	1962-09-28	700
ShareIss1Y	Investment	4.871	6.036	7.804	0.624	25.524	-13.625	-2.878	4.224	10.950	1927-07-30	1,122
ShareIss5Y	Investment	4.669	5.268	8.384	0.557	29.015	-8.052	-2.899	3.897	30.070	1931-07-31	1,074
ShareRepurchase	Investment	2.014	2.575	5.446	0.370	24.889	-8.318	-2.276	2.471	5.631	1972-07-31	582
ShareVol	Trading frictions	5.558	3.313	16.228	0.342	77.968	-29.050	-7.215	7.401	28.075	1926-09-30	1,123
ShortInterest	Trading frictions	9.553	5.975	11.068	0.863	20.305	-15.606	-4.439	5.620	16.318	1973-02-28	575
sinAlgo	Other	3.408	2.550	11.166	0.305	56.200	-15.830	-4.584	5.083	35.224	1951-03-31	838
Size	Other	4.371	3.084	13.765	0.318	52.279	-10.973	-4.235	5.913	53.260	1926-09-30	1,132
skew1	Trading frictions	5.853	4.086	7.151	0.818	18.390	-9.273	-2.320	3.658	7.431	1996-02-29	299
SmileSlope	Trading frictions	14.699	10.479	7.001	2.099	4.775	-4.653	-1.220	4.196	15.482	1996-02-29	299
SP	Value vs. growth	8.269	5.123	13.457	0.615	58.840	-25.006	-4.670	6.493	20.620	1951-07-31	834
SP_q	Value vs. growth	12.761	6.168	15.935	0.801	66.272	-36.687	-4.668	7.546	30.010	1961-09-29	712
Spinoff	Other	3.306	2.264	14.184	0.233	62.055	-20.934	-4.773	5.265	54.375	1926-09-30	1,132
std_turn	Trading frictions	6.164	3.007	19.765	0.312	80.934	-45.882	-8.412	8.973	25.169	1928-01-31	1,116
STreversal	Other	35.197	14.215	24.049	1.464	50.364	-36.964	-4.492	13.909	79.534	1926-09-30	1,132
SurpriseRD	Intangibles	1.044	1.412	6.136	0.170	49.918	-10.417	-2.360	2.650	16.462	1952-03-31	826
tang	Intangibles	4.304	3.219	11.188	0.385	37.320	-12.065	-4.163	4.874	38.744	1951-01-31	840
tang_q	Intangibles	6.218	4.622	9.497	0.655	52.453	-9.233	-3.559	4.753	27.608	1971-03-31	598
Tax	Profitability	4.278	5.236	6.812	0.628	34.392	-16.421	-2.321	3.227	11.110	1951-07-31	834
Tax_q	Profitability	0.871	0.908	7.389	0.118	65.873	-11.265	-2.793	2.401	32.551	1961-09-29	712
TotalAccruals	Investment	3.551	3.563	8.247	0.431	43.768	-7.858	-2.547	3.703	16.382	1952-07-31	822
UpRecomm	Intangibles	4.039	5.409	3.886	1.039	8.024	-6.836	-1.074	2.110	4.534	1993-12-31	325
VarCF	Other	-5.451	-2.710	16.525	-0.330	99.479	-30.941	-7.825	6.407	14.029	1953-07-31	810
VolMkt	Trading frictions	3.348	1.806	17.930	0.187	80.691	-31.954	-7.761	8.756	21.094	1927-07-30	1,122
VolSD	Trading frictions	3.516	2.388	14.200	0.248	39.969	-31.742	-5.564	6.072	43.474	1928-01-31	1,116
VolumeTrend	Other	6.617	5.230	12.168	0.544	29.105	-25.261	-3.705	5.224	45.626	1928-07-31	1,110
WW	Other	3.510	2.124	13.726	0.256	61.703	-16.077	-4.858	6.290	31.135	1952-01-31	828
WW_Q	Other	4.345	1.460	21.063	0.206	77.356	-21.674	-7.394	10.386	42.262	1970-11-30	601
XFIN	Investment	11.679	4.836	16.817	0.694	61.192	-36.495	-5.990	8.208	24.596	1972-07-31	582
zerotrade	Trading frictions	6.161	3.221	18.578	0.332	46.739	-27.052	-7.226	7.957	67.172	1926-09-30	1,132
zerotradeAlt1	Trading frictions	6.743	3.700	17.667	0.382	56.420	-27.511	-6.616	8.324	54.367	1927-01-31	1,128
zerotradeAlt12	Trading frictions	4.992	3.337	14.497	0.344	46.510	-21.010	-5.205	6.398	58.400	1927-02-28	1,127
ZScore	Profitability	-0.120	-0.055	16.524	-0.007	90.674	-19.919	-6.548	7.211	32.341	1963-01-31	696
ZScore_q	Profitability	-3.014	-1.176	17.989	-0.168	95.687	-29.248	-8.667	6.516	21.465	1971-10-29	591

This table shows descriptive statistics for raw anomaly returns. Panel A shows average statistics for each economic category. Panel B displays individual anomaly statistics. The columns show the acronym, the economic category, the mean return, t-stat of that return, standard deviation, Sharpe ratio, maximum Drawdown, minimum and maximum return, 5 and 95 percentile return, the start of the sample and the number of observations, respectively. Table A.1 gives a brief description of the firm characteristics.

B Timing Signals

This section describes the details of our timing signals. For each factor i, timing signal j and time t we determine a scaling factor $w_{i,t}^{j}$. The timed factor returns $f_{i,t+1}^{j}$ in the subsequent period are obtained from untimed returns $f_{i,t+1}$ as $f_{i,t+1}^{j} = f_{i,t+1} \cdot w_{i,t}^{j}$. Table B.1 provides detailed information about each timing signal. The columns show the acronym, the trading signal class, the original study, the corresponding journal, the original signals' definition and the definition of the scaling factor $w_{i,t}^{j}$ applied in our paper, respectively.

Table B.1: Summary of Timing Signals

Acronym	Category	Related literature	Implementation to obtain $w_{i,t}^{j}$ in our paper
MOM1	Momentum	Gupta and Kelly (2019)	Annualized return in month t scaled by annualized past return volatility over 3Y, capped at ± 2.
MOM2	Momentum	Gupta and Kelly (2019)	Annualized average return over the formation period $t-2$ to t, scaled by annualized past return volatility over 3Y, capped at ± 2.
MOM3	Momentum	Gupta and Kelly (2019)	Annualized average return over the formation period $t-5$ to t, scaled by annualized past return volatility over 3Y, capped at ± 2.
MOM4	Momentum	Gupta and Kelly (2019)	Annualized average return over the formation period $t-11$ to t, scaled by annualized past return volatility over 10Y, capped at ± 2.
MOM5	Momentum	Gupta and Kelly (2019)	Annualized average return over the formation period $t-35$ to t, scaled by annualized past return volatility over 10Y, capped at ± 2.
MOM6	Momentum	Gupta and Kelly (2019)	Annualized average return over the formation period $t-59$ to t, scaled by annualized past return volatility over 10Y, capped at ± 2.
MOM7	Momentum	Gupta and Kelly (2019)	Annualized average return over the formation period $t-11$ to $t-1$, scaled by annualized past return volatility over 3Y, capped at ± 2.
MOM8	Momentum	Gupta and Kelly (2019)	Annualized average return over the formation period $t-59$ to $t-12$, scaled by annualized past return volatility over 10Y, capped at ± 2.
MOM9	Momentum	Ehsani, Linnainmaa (2019)	Sign of return in month t.
MOM10	Momentum	Ehsani, Linnainmaa (2019)	Sign of cumulative return over months $t-2$ to t.
MOM11	Momentum	Ehsani, Linnainmaa (2019)	Sign of cumulative return over months $t-5$ to t.
MOM12	Momentum	Ehsani, Linnainmaa (2019)	Sign of cumulative return over months $t-11$ to t.
VOL1	Volatility	Moreira and Muir (2017)	Inverse of the variance of daily returns measured in month t, scaled by the average of all monthly variances of daily returns until t.
VOL2	Volatility	Moreira and Muir (2017)	Inverse of the standard deviation of daily returns measured in month t, scaled by the average of all monthly standard deviations of daily returns until t.
VOL3	Volatility	Moreira and Muir (2017)	Inverse of the variance of daily returns measured in month t, estimated from an AR(1) process for log variance, scaled by the average of all monthly variances of daily returns until t.
VOL4	Volatility	Cederburg, O'Doherty, Wang, Yan (2020)	Inverse of the variance of daily returns measured in month t, multiplied by 22 divided by the number of trading days in the month, scaled by the average of all monthly variances of daily returns until t.
VOL5	Volatility	DeMiguel, Utrera and Uppal (2021)	Inverse of the standard deviation of daily market returns measured in month t, scaled by the average of all monthly standard deviations of daily returns until t.
VOL6	Volatility	Reschenhofer and Zechner (2021)	Level of implied volatility (CBOE VIX index) in month t.
VOL7	Volatility	Reschenhofer and Zechner (2021)	Level of implied skewness (CBOE SKEW index) in month t.
REV1	Reversal	Moskowitz, Ooi, and Pedersen (2012)	1 minus annualized average return over the period $t-59$ to t.
REV2	Reversal	Moskowitz, Ooi, and Pedersen (2012)	1 minus annualized average return over the period $t-119$ to t.

Table B.1 – cont.

Acronym	Category	Related literature	Implementation in our paper
TSMOM1	Momentum	Moskowitz, Ooi, and Pedersen (2012)	Sign of return in month t, multiplied by 40% divided by ex-ante volatility, where ex-ante volatility is the square root of exponentially weighted moving average of squared daily returns.
TSMOM2	Momentum	Moskowitz, Ooi, and Pedersen (2012)	Sign of cumulative return over months $t-2$ to t, multiplied by 40% divided by ex-ante volatility, where ex-ante volatility is the square root of exponentially weighted moving average of squared daily returns.
TSMOM3	Momentum	Moskowitz, Ooi, and Pedersen (2012)	Sign of cumulative return over months $t-5$ to t, multiplied by 40% divided by ex-ante volatility, where ex-ante volatility is the square root of exponentially weighted moving average of squared daily returns.
TSMOM4	Momentum	Moskowitz, Ooi, and Pedersen (2012)	Sign of cumulative return over months $t-11$ to t, multiplied by 40% divided by ex-ante volatility, where ex-ante volatility is the square root of exponentially weighted moving average of squared daily returns.
VAL1	Valuation	Campbell and Shiller (1988), Cohen, Polk, and Vuolteenaho (2003), Haddad, Kozak, and Santosh (2020)	We first calculate the BTM spread as the difference of log book-to-market ratio of long minus short leg. The signal is obtained as the difference of the BTM spread at time t minus the expanding mean BTM spread up to time t, scaled by the standard deviation of the difference.
VAL2	Valuation	Campbell and Shiller (1988), Cohen, Polk, and Vuolteenaho (2003), Haddad, Kozak, and Santosh (2020)	We first calculate the BTM spread as the difference of log book-to-market ratio of long minus short leg. The signal is obtained as the difference of the BTM spread at time t minus the 5 year rolling mean BTM spread up to time t, scaled by the standard deviation of the difference.
VAL3	Valuation	Campbell and Shiller (1988), Cohen, Polk, and Vuolteenaho (2003), Haddad, Kozak, and Santosh (2020)	We first calculate the BTM spread as the difference of book-to-market ratio of long minus short leg using the book-value of December of last year. The signal is obtained as the difference of the BTM spread at time t minus the expanding mean BTM spread up to time t, scaled by the standard deviation of the difference.
VAL4	Valuation	Campbell and Shiller (1988), Cohen, Polk, and Vuolteenaho (2003), Haddad, Kozak, and Santosh (2020)	We first calculate the BTM spread as the difference of book-to-market ratio of long minus short leg using the book-value of December of last year. The signal is obtained as the difference of the BTM spread at time t minus the 5 year rolling mean BTM spread up to time t, scaled by the standard deviation of the difference.
VAL5	Valuation	Campbell and Shiller (1988), Cohen, Polk, and Vuolteenaho (2003), Haddad, Kozak, and Santosh (2020)	We first calculate the BTM spread as the difference of log book-to-market ratio of long minus short leg using quarterly book equity over market equity. The signal is obtained as the difference of the BTM spread at time t minus the expanding mean BTM spread up to time t, scaled by the standard deviation of the difference.
VAL6	Valuation	Campbell and Shiller (1988), Cohen, Polk, and Vuolteenaho (2003), Haddad, Kozak, and Santosh (2020)	We first calculate the BTM spread as the difference of log book-to-market ratio of long minus short leg using quarterly book equity over market equity. The signal is obtained as the difference of the BTM spread at time t minus the 5 year rolling mean BTM spread up to time t, scaled by the standard deviation of the difference.
SPREAD1	Characteristic spread	Huang, Liu, Ma, Osiol (2011)	We first calculate the characteristic spread as the difference of long minus short leg. The signal is obtained as the difference of the characteristic spread at time t minus the expanding mean characteristic spread up to time t, scaled by the standard deviation of the difference.
SPREAD2	Characteristic spread	Huang, Liu, Ma, Osiol (2011)	We first calculate the characteristic spread as the difference of long minus short leg. The signal is obtained as the difference of the characteristic spread at time t minus the 5 year rolling mean characteristic spread up to time t, scaled by the standard deviation of the difference.
IPS1	Issuer-purchaser spread	Greenwood and Hanson (2012)	We first calculate the difference of the average year-on-year change in net stock issuance for net equity issuers versus repurchasers, where net stock issuance is the change in log split-adjusted shares outstanding from Compustat (CSHO $\times$ AJEX). Then we calculate the standard deviation of the difference. The signal is obtained as the difference of the spread at time t minus the expanding mean up to time t, scaled by the standard deviation of the difference.

Table B.1 – cont.

Acronym	Category	Related literature	Implementation in our paper
IPS2	Issuer-purchaser spread	Greenwood and Hanson (2012)	We first calculate the difference of the average year-on-year change in net stock issuance for net equity issuers versus repurchasers, where net stock issuance is the change in log split-adjusted shares outstanding from Compustat (CSHO $\times$ AJEX). Then we calculate the standard deviation of the difference. The signal is obtained as the difference of the spread at time t minus the 5 year rolling mean up to time t, scaled by the standard deviation of the difference.
IPS3	Issuer-purchaser spread	Pontiff and Woodgate (2008)	We first calculate the difference of the average growth in the numbers of shares between $t-18$ and $t-6$ for net equity issuers versus repurchasers, where the number of shares is calculated as shrout/cfacshr to adjust for splits from CRSP. Then we calculate the standard deviation of the difference. The signal is obtained as the difference of the spread at time t minus the expanding mean up to time t, scaled by the standard deviation of the difference.
IPS4	Issuer-purchaser spread	Pontiff and Woodgate (2008)	We first calculate the difference of the average growth in the numbers of shares between $t-18$ and $t-6$ for net equity issuers versus repurchasers, where the number of shares is calculated as shrout/cfacshr to adjust for splits from CRSP. Then we calculate the standard deviation of the difference. The signal is obtained as the difference of the spread at time t minus the 5 year rolling mean up to time t, scaled by the standard deviation of the difference.
IPS5	Issuer-purchaser spread	Bradshaw, Richardson, Sloan (2006)	We first calculate the difference of the average sale of common stock (sstk) minus purchase of common stock (prstkc), scaled by average total assets (at), for net equity issuers versus repurchasers. We exclude stocks where the absolute value of ratio is greater than 1. Then we calculate the standard deviation of the difference. The signal is obtained as the difference of the spread at time t minus the expanding mean up to time t, scaled by the standard deviation of the difference.
IPS6	Issuer-purchaser spread	Bradshaw, Richardson, Sloan (2006)	We first calculate the difference of the average sale of common stock (sstk) minus purchase of common stock (prstkc), scaled by average total assets (at), for net equity issuers versus repurchasers. We exclude stocks where the absolute value of ratio is greater than 1. Then we calculate the standard deviation of the difference. The signal is obtained as the difference of the spread at time t minus the 5 year rolling mean up to time t, scaled by the standard deviation of the difference.

C Additional Results

Table C.1: Performance Impact of Factor Timing with Single Signals

This table shows timing success of different signals for individual factors, grouped into economic categories. It is analogous to Table 1 in the main text, but shows results for additional signal categories. N_f reports the number of factors within each category. The left part of the panel shows the alpha for each factor i and signal j against its raw (untimed) counterpart. Alpha is obtained as the intercept in the following regression: $f^j_{i,t+1} = \alpha_{i,j} + \beta_{i,j} f_{i,t+1} + \epsilon_{t+1}$. α, $\alpha > 0$, and $\alpha < 0$ present the average alpha, and the number of factors with a positive and negative α, respectively. We report average t-statistics and the number of significant factors at the 5 percent level in brackets, based on heteroscedasticity-adjusted standard errors. The right part shows the average difference in the annualized Sharpe ratio of the timed versus untimed factor across factor/signal combinations. For Sharpe ratios, we use the z-statistic from the Jobson and Korkie (1981) test of the null that $SR(f^j_i - f_i) = 0$. Panel A report results for the characteristic spreads, Panel B report results for the Issuer-purchaser spread, Panel C for the Reversal signals and Panel D for the valuation spread signals. We describe the factors and their allocation to an economic category in Table A.1. Table B.1 describes the timing signals.

		Time series regression			Sharpe ratio difference		
	N_f	α	$\alpha > 0$	$\alpha < 0$	ΔSR	$\Delta SR > 0$	$\Delta SR < 0$
A. Characteristic spread							
All factors	318	-0.628 [-0.223]	134 [18]	184 [31]	-0.352 [-2.126]	53 [14]	265 [167]
Intangibles	53	-0.152 [-0.235]	21 [2]	32 [4]	-0.338 [-1.792]	6 [1]	47 [22]
Investment	46	-0.360 [-0.303]	17 [3]	29 [2]	-0.394 [-2.452]	8 [5]	38 [30]
Momentum	22	-1.302 [-0.001]	10 [2]	12 [2]	-0.691 [-4.166]	1 [0]	21 [15]
Profitability	35	-0.649 [-0.010]	20 [0]	15 [2]	-0.274 [-1.643]	8 [0]	27 [15]
Trading frictions	46	0.170 [0.064]	22 [4]	24 [2]	-0.206 [-1.277]	12 [4]	34 [19]
Value vs. growth	41	-1.805 [-0.615]	14 [4]	26 [8]	-0.374 [-2.556]	5 [2]	36 [25]
Other	75	-0.767 [-0.293]	29 [4]	46 [8]	-0.352 [-2.077]	14 [2]	62 [40]
B. Issuer-purchaser spread							
All factors	318	1.389 [0.507]	209 [38]	109 [8]	-0.329 [-1.599]	69 [18]	249 [134]
Intangibles	53	1.045 [0.432]	35 [5]	18 [1]	-0.324 [-1.477]	12 [2]	42 [20]
Investment	46	0.508 [0.226]	26 [4]	20 [2]	-0.454 [-2.268]	6 [5]	40 [28]
Momentum	22	0.864 [0.448]	13 [2]	9 [0]	-0.723 [-3.567]	1 [0]	21 [16]
Profitability	35	1.415 [0.649]	23 [6]	12 [1]	-0.250 [-1.255]	10 [2]	25 [12]
Trading frictions	46	2.322 [0.492]	30 [6]	16 [1]	-0.216 [-0.945]	14 [4]	32 [14]
Value vs. growth	41	1.574 [0.704]	29 [7]	12 [1]	-0.305 [-1.590]	8 [3]	33 [16]
Other	75	1.642 [0.586]	53 [8]	22 [1]	-0.261 [-1.266]	19 [3]	56 [28]
C. Reversal							
All factors	318	0.005 [-0.156]	150 [13]	168 [29]	-0.005 [-0.301]	142 [13]	176 [42]
Intangibles	53	-0.058 [-0.329]	20 [1]	33 [2]	-0.008 [-0.420]	20 [1]	32 [5]
Investment	46	-0.058 [-0.329]	20 [1]	33 [2]	-0.008 [-0.420]	20 [1]	32 [5]
Momentum	22	0.014 [-0.131]	10 [2]	12 [3]	-0.012 [-0.516]	9 [3]	13 [5]
Profitability	35	0.165 [-0.057]	17 [2]	18 [1]	0.000 [-0.283]	15 [2]	20 [4]
Trading frictions	46	0.049 [0.460]	30 [4]	16 [0]	0.003 [0.432]	30 [2]	16 [0]
Value vs. growth	41	-0.091 [-0.549]	16 [0]	26 [7]	-0.012 [-0.704]	15 [2]	26 [10]
Other	75	0.012 [-0.044]	39 [2]	36 [6]	-0.006 [-0.212]	36 [2]	38 [10]
D. Valuation							
All factors	318	0.898 [0.331]	191 [32]	127 [12]	-0.388 [-1.931]	57 [14]	261 [148]
Intangibles	53	0.658 [0.228]	30 [4]	23 [2]	-0.401 [-1.811]	10 [1]	43 [23]
Investment	46	0.446 [0.269]	27 [4]	19 [1]	-0.487 [-2.452]	8 [4]	38 [30]
Momentum	22	4.055 [1.306]	17 [7]	5 [0]	-0.842 [-3.795]	0 [0]	22 [16]
Profitability	35	1.503 [0.588]	24 [5]	11 [1]	-0.313 [-1.535]	7 [1]	28 [14]
Trading frictions	46	1.150 [0.346]	29 [4]	17 [1]	-0.223 [-1.040]	13 [3]	33 [13]
Value vs. growth	41	-1.164 [-0.397]	16 [2]	25 [6]	-0.405 [-2.514]	4 [2]	37 [23]
Other	75	1.108 [0.423]	47 [6]	28 [1]	-0.312 [-1.562]	15 [3]	60 [30]

Table C.2: Stock-level Timing Portfolios (sub periods)

This table shows a sub sample analysis for the stock-level timing portfolios presented for the full sample in Table 9 in the main text. Results are shown for long-only equity portfolios. To this end, we aggregate the underlying security weights from all timed factor portfolios. We then retain only firms that have positive total weights. All subsamples are then again split by large (above the NYSE median market cap) and small firms (below the NYSE median market cap). ALL_VW is the value-weighted portfolio return of a small and large cap stocks respectively. Untimed refers to portfolio weights based on the original factor definition. PLS 1 timed shows portfolio timing based on partial least squares regressions with a single component. We report annualized mean return (R), standard deviation (SD), Sharpe ratio (SR), maximum drawdown (maxDD), average number of firms in the portfolio (N), and annualized turnover (Turn). We describe the factors and their allocation into an economic category in Table A.1.

	R	SD	SR	maxDD	N	Turn
01/1974 – 12/1989						
A. Small capitalization stocks						
ALL_VW	15.546	20.520	0.375	37.069	4,096	6.211
Untimed	25.598	22.587	0.786	33.573	2,329	343.044
PLS 1 timed	26.066	22.489	0.810	34.720	2,320	419.587
B. Large capitalization stocks						
ALL_VW	9.438	16.749	0.095	36.349	826	3.061
Untimed	11.652	17.079	0.222	38.021	242	382.965
PLS 1 timed	14.125	18.660	0.336	38.762	273	505.542
01/1990 – 12/2004						
C. Small capitalization stocks						
ALL_VW	12.923	20.058	0.441	36.403	4,860	8.014
Untimed	31.066	19.871	1.358	25.512	2,721	252.879
PLS 1 timed	34.626	23.916	1.278	36.895	2,609	367.029
D. Large capitalization stocks						
ALL_VW1	9.498	14.860	0.365	46.851	1,049	3.962
ORG	14.213	14.381	0.705	26.514	377	260.898
PLS 1 timed	15.184	16.501	0.673	41.738	403	416.474
01/2005 – 12/2020						
E. Small capitalization stocks						
ALL_VW	10.019	20.730	0.425	55.076	2,937	6.999
Untimed	16.120	22.276	0.669	57.828	1,629	260.989
PLS 1 timed	16.701	21.428	0.723	50.035	1,492	416.863
F. Large capitalization stocks						
ALL_VW	8.872	14.966	0.512	51.585	920	3.462
Untimed	10.547	16.786	0.556	49.111	406	290.840
PLS 1 timed	11.310	16.852	0.599	51.084	457	440.629

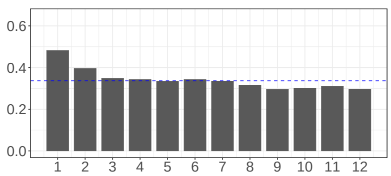
Table C.3: Stock-level Timing Portfolios: Best-in-class

This table shows variants of the stock-level timing portfolios presented in Table 9 in the main text. Results are shown for long-only equity portfolios. To this end, we aggregate the underlying security weights from all timed factor portfolios. We then retain only firms that have positive total weights. In contrast to Table 9 in the main text, we build more concentrated portfolios, by focusing only on the 20% (50%) with the largest positive aggregate weights. Panels A and B report results for small and large-capitalization stocks in the CRSP universe, where we split the sample in June of year t using the median NYSE market equity and keep firms from July of year t to June of year $t+1$. ALL_VW is the value-weighted portfolio return of a small and large cap stocks respectively. Untimed refers to portfolio weights based on the original factor definition. PLS 1 timed shows portfolio timing based on partial least squares regressions with a single component. We report annualized mean return (R), standard deviation (SD), Sharpe ratio (SR), maximum drawdown (maxDD), average number of firms in the portfolio (N), and annualized turnover (Turn). The sample period is January 1974 to December 2020. We describe the factors and their allocation into an economic category in Table A.1.

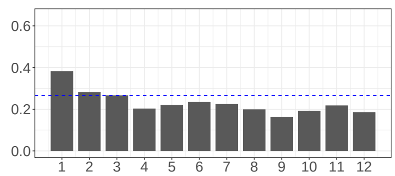
	R	SD	SR	maxDD	N	Turn
A. Small capitalization stocks						
ALL_VW	12.832	20.420	0.413	55.076	3,945	7.053
Untimed | w in top 50%	25.286	21.985	0.950	57.150	1,108	205.429
Untimed | w in top 20%	26.701	22.619	0.986	58.616	444	177.309
PLS 1 timed | w in top 50%	26.981	23.291	0.970	49.481	1,065	262.895
PLS 1 timed | w in top 20%	28.243	24.755	0.964	50.888	427	220.542
B. Large capitalization stocks						
ALL_VW	9.265	15.538	0.314	51.585	929	3.484
Untimed | w in top 50%	12.088	16.073	0.479	48.143	171	234.445
Untimed | w in top 20%	11.571	15.865	0.453	45.738	69	208.445
PLS 1 timed | w in top 50%	13.565	17.484	0.525	51.669	189	340.172
PLS 1 timed | w in top 20%	13.130	17.724	0.493	51.665	76	316.498

Figure C.1: Timing persistence

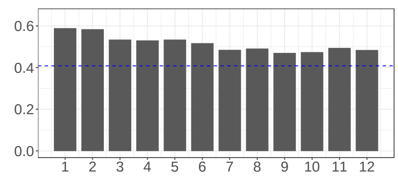
This figure illustrates the Sharpe ratios of univariate factor timing portfolios and their respective durations of predictive accuracy. For each factor i, the predicted returns $\hat{f}_{i,t+1}$ are from partial least squares (PLS) regressions, where the number of components equals 1. The timed portfolio at time T, denoted by $f^{\tau}_{i,t+T}$, is equal to the original factor return, when the forecast is positive. In the event of a negative predicted return, the timed factor return is essentially its inverse, i.e. the short-long untimed portfolio return. T, with $T \in \{1, ..., 12\}$, denotes the lagged time (in months) between the PLS prediction and the actual investment. All statistics are obtained in a two-step procedure: First, we compute statistics for each individual factor separately for its out-of-sample period. Second, we take cross-sectional averages. The blue line represents the average Sharpe ratio of all untimed factors. We describe the factors and their allocation to an economic category in Table A.1. Table B.1 describes the timing signals.



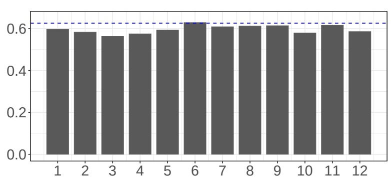
(a) All factors



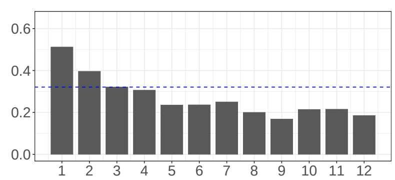
(b) Intangibles



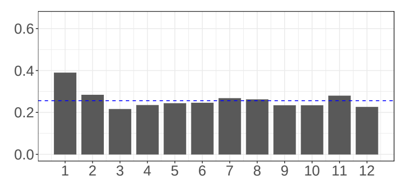
(c) Investment



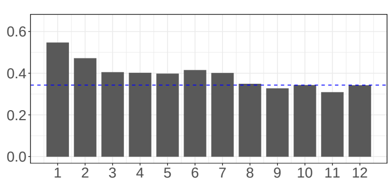
(d) Momentum



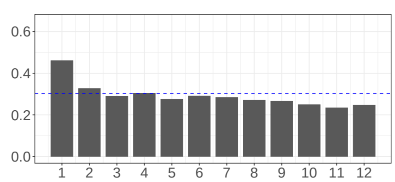
(e) Profitability



(f) Trading frictions



(g) Value vs. growth



(h) Other